Comparative Benchmark of State-of-Charge Observers and Physics-Informed Neural Networks Across Battery Aging and Thermal Regimes

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*Although battery State-of-Charge (SOC) estimation has been extensively investigated, existing comparative studies predominantly evaluate fresh cells at room temperature, obscuring algorithmic vulnerabilities under operational aging and thermal extremes. This work establishes a dual-tier benchmark of six representative nonlinear observers, deep recurrent networks, and physics-informed architectures (EKF, AEKF, DEKF, SMO, BiLSTM-Attn, and PINN-SOC) combining direct experimental cycler datasets with calibrated electro-thermal-aging co-simulation. Methodologies are systematically assessed across progressive capacity fade (*100% → 80% *SOH), wide ambient temperatures (*0°C *to* 45°C*), flat-plateau* $LiFePO_4$ *thermodynamics, and 8S series pack inconsistency. Results reveal that whereas uncompensated EKF suffers a 45.7-fold RMSE escalation (*0.065% *to* 2.973%*) as interfacial impedance expands, online parameter-tracking DEKF confines lifespan RMSE to* 0.585%*. Furthermore, we formulate a hybrid Physics-Informed Dual Kalman Filter (PIDKF) that reveals chemistry-dependent complementarity—neural overpotential compensation dominates on sloping-OCV NMC (*0.414% *RMSE), whereas dynamic hysteresis tracking dominates on flat-plateau LFP (*0.149% *RMSE) under calibrated co-simulation for the embedded exogenous-Jacobian configuration (*0.412% *and* 0.148% *for exact Jacobian in Table 9). Finally, bare-metal profiling on ARM Cortex-M microcontrollers (*14.20 µs *measured on Cortex-M4;* 4.97 µs *constant-IPC clock-scaled estimate on Cortex-M7) delineates the Pareto efficiency frontier for industrial battery management systems.*

Keywords: State-of-charge, Lithium-ion battery, Battery degradation, Dual Extended Kalman filter, Physics-informed neural network, Electrochemical impedance spectroscopy, Embedded BMS, ASME JEECS

Nomenclature

$\mathbf{A}_k, \mathbf{B}_k, \mathbf{g}_k$ = discrete state transition, input, and hysteresis vectors
C_1, C_2, Q_{cpe} = polarization capacitances (F) and CPE parameter ($\text{F} \cdot \text{s}^{\alpha-1}$)
$\mathbf{C}_{\theta,k}, \mathbf{H}_k$ = parameter sensitivity and observation Jacobian matrices
E_a, R = Arrhenius activation energy ($\text{J} \cdot \text{mol}^{-1}$) and gas constant
$I_k, I(t)$ = cell load current at time step k, positive for discharge (A)
$\mathbf{K}_k, \mathbf{L}$ = Kalman filter gain and sliding mode feedback gain
$M(z), V_h$ = maximum dynamic hysteresis function and hysteresis state (V)
$\mathbf{P}_k, \mathbf{Q}_k, \mathbf{R}_k$ = state error, process, and measurement covariance matrices
$Q_{\text{act}}, Q_{\text{nom}}$ = actual degraded and nominal pristine capacities ($\text{A} \cdot \text{h}$)
R_0, R_1, R_2 = effective series resistance ($R_{\text{ohm}} + R_{\text{sei}}$) and polarization resistances (Ω)
$R_{\text{ct}}, R_{\text{sei}}$ = charge-transfer and SEI layer resistances from EIS (Ω)
$S, \mathbf{x}_k, \mathbf{y}_k$ = sliding manifold, state vector, and observation variable
T, T_{amb} = battery core temperature and ambient temperature (K or °C)
$U_1, U_2, \delta V_{\text{pol}}$ = RC overpotentials and unmodeled residual compensation (V)
V_t, V_k, V_{ocv} = terminal and thermodynamic open-circuit voltages (V)
$\mathbf{w}_k, v_k$ = process and measurement noise signals
$Z(j\omega), Z_w$ = electrochemical impedance and Warburg diffusion impedance (Ω)

Greek Letters

α, σ_w = constant-phase element exponent and Warburg coefficient
$\beta, \lambda_{\text{SH}}$ = hysteresis decay rate and Sage-Husa forgetting factor
$\gamma_{\min}, \eta_s, \Phi$ = sliding mode injection gain, margin, and disturbance bound
$\Delta t, \mu$ = discrete sampling interval (1.0 s) and boundary thickness (V)
η, κ = Faradaic coulombic efficiency and condition number
κ_0, κ_1 = Butler-Volmer overpotential bound coefficients
τ_1, τ_2 = electrochemical polarization time constants ($\tau_i = R_i C_i$, s)
$\boldsymbol{\theta}_k$ = parameter state vector in Dual Kalman Filter

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$$[R_{0,k}, 1/Q_{\text{act},k}]^T$$

Subscripts and Superscripts

act, nom, eff = actual degraded, nominal pristine, and effective states
k, meas, T = temporal index, experimental measurement, and transpose operator

1 Introduction

The global transition toward carbon-neutral transportation and renewable grid energy storage has accelerated the deployment of high-energy-density lithium-ion battery (LIB) packs [1–3]. In automotive battery management systems (BMS), the State-of-Charge (SOC) represents the ratio of currently available capacity to maximum usable capacity. SOC serves as the fundamental state metric required for accurate driving range estimation, optimal regenerative braking control, prevention of over-charge and over-discharge, and dynamic power capability (State-of-Power, SOP) calculation [4–6].

Unlike terminal voltage or operating temperature, the SOC cannot be directly measured via physical sensors during vehicle operation due to complex internal solid-state diffusion, interfacial charge-transfer kinetics, and structural phase transitions occurring within the porous insertion electrodes [7,8]. Consequently, online SOC estimation must rely on mathematical observers that interpret terminal current, voltage, and surface temperature signals [9–11].

1.1 Taxonomy of Algorithmic Estimation Paradigms. Over the past two decades, battery state estimation methodologies in the electrochemical control literature have evolved into five primary algorithmic paradigms [5,12]:

1. *Direct Empirical Integration*: Coulomb counting provides an $O(1)$ baseline [4] but accumulates open-loop drift from sensor bias, ADC quantization, and parasitic side reactions. Static OCV lookup achieves $< 1\%$ accuracy yet requires prolonged cell relaxation (30–120 min) to extinguish diffusion overpotentials, precluding active driving deployment [13,14].

2. *Model-Based Nonlinear Observers*: Equivalent circuit models (1-RC Thevenin, 2-RC Dual Polarization) coupled with recursive filtering enable closed-loop state tracking [10,15]. The Extended Kalman Filter (EKF) linearizes OCV via first-order Taylor expansion [16,17], whereas Adaptive EKF (AEKF) updates noise covariances via Sage-Husa recursions [18,19]. Sigma-point filters (UKF, CKF) propagate deterministic sigma points for higher-order accuracy [20–22]. To counter aging drift, Dual EKF (DEKF) co-estimates fast states and slow parameters (R_0, Q_{act}) [23,24], while Sliding Mode Observers (SMO) drive error trajectories onto a switching manifold with bounded disturbance rejection [25–27].

3. *Physics-Based Electrochemical Observers*: Doyle-Fuller-Newman (P2D) models resolve solid/liquid lithium diffusion, potential fields, and Butler-Volmer kinetics [7,8,28]. Reduced-order formulations (SPM, SPMe) facilitate PDE observer design [29–31], yet microcontroller deployment remains constrained by heavy spatial discretization burdens.

4. *Pure Data-Driven Machine Learning*: Deep recurrent architectures (LSTM, BiLSTM, GRU, TCN) and self-attention Transformers capture complex temporal load dynamics [19,32–38]. Nevertheless, black-box networks lack physical invariance and suffer severe accuracy degradation when evaluated outside their training distribution.

5. *Physics-Informed Machine Learning*: Physics-Informed Neural Networks (PINNs) embed differential-algebraic conservation laws (Faradaic species conservation, KVL, thermodynamic monotonicity) directly into backpropagation loss [39–41]. Evaluating physical residuals via automatic differentiation across dense collocation domains constrains network manifolds to physically admissible trajectories, mitigating out-of-distribution divergence.

1.2 Critical Review of Existing Benchmark Studies and Knowledge Gaps. Despite extensive research on SOC estimation, existing comparative benchmarks exhibit five fundamental deficiencies:

Gap 1: Pristine Cell Bias and Degradation Neglect. Most benchmarks evaluate fresh cells (SOH = 100%) at beginning-of-life [10,33,42]; uncompensated observer vulnerability to progressive SEI growth down to 80% SOH remains unquantified.

Gap 2: Decoupling of Thermal Kinetics from Observers. Overpotentials scale exponentially with Arrhenius temperature dynamics [11,15], yet benchmarks predominantly test at 25°C, leaving thermal extremes (0°C, 45°C) uncharacterized.

Gap 3: Observability Degradation on Flat Plateaus. Whereas NMC exhibits steep OCV-SOC gradients, $LiFePO_4$ (LFP) cells feature flat voltage plateaus ($d\text{OCV}/d\text{SOC} \approx 0$) and pronounced dynamic hysteresis [13,14,43]. Although mechanical force fusion mitigates this deficit [44], purely electrical tracking without auxiliary strain sensors remains unbenchmarked.

Gap 4: Multi-Source Sensor Corruption Vulnerability. Current transducers suffer zero-point thermal bias drift, while voltage sensing lines experience inverter PWM electromagnetic interference (EMI) [45,46]. Comparative studies rarely evaluate simultaneous drift and EMI stress.

Gap 5: Embedded Hardware Disconnect and Pack Inconsistency. Complex observers are typically evaluated on desktop workstations [35,47]. Cycle-accurate profiling on automotive microcontrollers (ARM Cortex-M4/M7) and scaling to series packs under cell inconsistency are virtually absent [24,46].

1.3 Contributions of This Work. To address these open challenges, this paper establishes a dual-tier benchmark of six foundational and modern state estimation methodologies across ten operational stress phases. The primary contributions are fivefold:

1. *Multi-Attribute Evaluation Across 10 Benchmark Phases*: We benchmark EKF, AEKF, DEKF, SMO, BiLSTM-Attention, and PINN-SOC across standardized driving schedules (UDDS, US06, DST, FUDS), 5 progressive aging stages (100%–80% SOH), 3 ambient temperatures (0°C, 25°C, and 45°C), large initial state uncertainty ($\Delta\text{SOC}_0 = -40\%$), multi-attribute radar synthesis, LFP flat plateaus, sensor bias/EMI corruptions, model-based EIS frequency analysis, 8S series pack inconsistency, and embedded microcontroller profiling.

2. *Formulation of the Aging Sensitivity Factor (ASF)*: A quantitative Aging Sensitivity Factor is introduced, revealing a 45.7-fold RMSE escalation in standard EKF as SOH fades to 80%, whereas online parameter-tracking DEKF strictly bounds lifespan error growth.

3. *Electrochemical Impedance Spectroscopy (EIS) Frequency-Domain Bridge*: Time-domain observer tracking drift is correlated with model-based Randles–CPE Nyquist impedance spectra (100 kHz to 1 mHz), illustrating how charge-transfer impedance growth amplifies polarization mismatch.

4. *Chemistry-Dependent Hybrid PIDKF Architecture*: A lightweight hybrid observer is formulated by coupling DEKF parameter co-estimation with dynamic hysteresis tracking and physics-informed neural polarization compensation, demonstrating that neural compensation dominates on aged NMC (0.414% RMSE) while hysteresis tracking dominates on aged LFP (0.149% RMSE) under calibrated co-simulation.

5. *Embedded Profiling on ARM Cortex-M Architectures*: Cycle-accurate execution complexity is measured on ARM Cortex-M4 (‘DWT->CYCCNT‘ at 168 MHz) and clock-scaled to Cortex-M7 (480 MHz) via CMSIS-DSP, delineating the empirical Pareto frontier and establishing practical BMS selection rules.

2 Electrochemical Battery Modeling and Degradation Dynamics

2.1 Second-Order Dual Polarization (2-RC) Model Formulation. The electrical behavior of the lithium-ion cell is rep-

resented by a second-order Dual Polarization equivalent circuit model, widely acknowledged as the optimal balance between mathematical tractability and dynamic fidelity [10,48]. The circuit comprises an open-circuit voltage source $V_{\mathrm{ocv}}(\mathrm{SOC})$, an ohmic series resistance R_0, and two parallel resistance-capacitance ($R_i \parallel C_i$) networks representing fast charge-transfer interfacial kinetics and slow solid-state lithium diffusion processes, respectively:

$$\frac{d\,\mathrm{SOC}(t)}{dt} = -\frac{\eta(T)}{3600\,Q_{\mathrm{act}}}\,I(t), \tag{1}$$

$$\frac{dU_1(t)}{dt} = -\frac{U_1(t)}{R_1C_1} + \frac{I(t)}{C_1}, \quad \frac{dU_2(t)}{dt} = -\frac{U_2(t)}{R_2C_2} + \frac{I(t)}{C_2} \tag{2}$$

where $I(t)$ denotes load current (defined positive for discharge and negative for charge), $\eta(T, I)$ represents Faradaic coulombic efficiency ($\eta = 1$ during discharge $I \geq 0$, and $\eta \in (0, 1]$ during charge $I < 0$ to account for parasitic side reactions), and Q_{act} denotes actual usable capacity in ampere-hours (A·h). The terminal output voltage $V_t(t)$ satisfies Kirchhoff's voltage law:

$$V_t(t) = V_{\mathrm{ocv}}(\mathrm{SOC}(t)) - U_1(t) - U_2(t) - I(t)R_0. \tag{3}$$

Under discharge ($I(t) > 0$), overpotentials $U_i(t) \geq 0$ and ohmic drop $I(t)R_0 > 0$ subtract from thermodynamic potential, ensuring $V_t < V_{\mathrm{ocv}}$. Irreversible dissipation rate $(U_1 + U_2 + IR_0)I/T \geq 0$ strictly satisfies the Second Law of Thermodynamics.

Discretizing the continuous state equations with sampling interval $\Delta t = 1.0$ s via zero-order hold (ZOH) under the standard assumption that $I(t) = I_k$ remains constant over interval $t \in [t_k, t_{k+1})$ yields the discrete-time state-space representation. Evaluating the continuous convolution integral $\int_0^{\Delta t} \exp(-(\Delta t - s)/\tau_i)\frac{I_k}{C_i}\,ds = R_i[1 - \exp(-\Delta t/\tau_i)]I_k$ (noting $\tau_i = R_iC_i$) establishes the exact discrete-time equivalent:

$$\mathbf{x}_{k+1} = \mathbf{A}_k\mathbf{x}_k + \mathbf{B}_kI_k + \mathbf{w}_k, \tag{4}$$

where the state vector is $\mathbf{x}_k = [\mathrm{SOC}_k, U_{1,k}, U_{2,k}]^T$, the process noise vector is $\mathbf{w}_k \sim \mathcal{N}(0, \mathbf{Q}_k)$, and the system matrices are given by:

$$\mathbf{A}_k = \mathrm{diag}\left(1, e^{-\Delta t/\tau_1}, e^{-\Delta t/\tau_2}\right), \tag{5}$$

$$\mathbf{B}_k = \left[-\frac{\eta\Delta t}{3600\,Q_{\mathrm{act}}}, \quad R_1\left(1 - e^{-\frac{\Delta t}{\tau_1}}\right), \quad R_2\left(1 - e^{-\frac{\Delta t}{\tau_2}}\right)\right]^T, \tag{6}$$

with polarization time constants $\tau_1 = R_1C_1$ and $\tau_2 = R_2C_2$ (calibrated at 25°C to $\tau_1 = 24.0$ s, $\tau_2 = 112.5$ s for NMC; 37.5 s, 144.0 s for LFP). The nonlinear observation equation is:

$$y_k = V_k = V_{\mathrm{ocv}}(\mathrm{SOC}_k) - U_{1,k} - U_{2,k} - I_kR_0 + v_k, \tag{7}$$

where $v_k \sim \mathcal{N}(0, R_k)$ represents measurement noise.

2.2 Arrhenius Thermal-Kinetic Coupling. Electrochemical reaction rates and ionic transport obey Arrhenius kinetics [11,15]. Defining the temperature scaling factor $f_{T,i}(T) \equiv \exp[\frac{E_{a,i}}{R}(\frac{1}{T} - \frac{1}{T_{\mathrm{ref}}})]$ ($T_{\mathrm{ref}} = 298.15$ K, $R = 8.314$ J·mol^{-1}·K^{-1}, $E_{a,0} \approx 20.5$ kJ·mol^{-1}, $E_{a,1} \approx 28.4$ kJ·mol^{-1}), ohmic and polarization resistances scale via:

$$R_0(T) = R_{0,\mathrm{ref}}f_{T,0}(T), \tag{8}$$

$$R_i(T) = R_{i,\mathrm{ref}}f_{T,i}(T), \quad i \in \{1, 2\} \tag{9}$$

At 0°C, electrolyte viscosity rises, causing a 2.13-fold increase in R_0 and 2.85-fold increase in R_1 ($\sim 2.5\times$ total impedance surge via Eqs. (8)–(9)).

2.3 Semi-Empirical Aging Parameterization Across Degradation Regimes. While microscopic degradation stems from continuous SEI growth and particle cracking [11,49], for real-time estimation across thermal and aging regimes, State-of-Health is anchored at reference temperature $T_{\mathrm{ref}} = 25$°C (SOH $\equiv Q_{\mathrm{act}}(T_{\mathrm{ref}})/Q_{\mathrm{init}} \in [0.80, 1.00]$), while usable capacity and internal resistances follow separable formulations:

$$R_0(T, \mathrm{SOH}) = R_{0,\mathrm{nom}}f_{T,0}(T)[1.0 + 2.50\,(1.0 - \mathrm{SOH})], \tag{10}$$

$$R_1(T, \mathrm{SOH}) = R_{1,\mathrm{nom}}f_{T,1}(T)[1.0 + 7.50\,(1.0 - \mathrm{SOH})], \tag{11}$$

$$Q_{\mathrm{act}}(T, \mathrm{SOH}) = \mathrm{SOH} \cdot Q_{\mathrm{init}} \exp\left[-\frac{E_{a,Q}}{R}\left(\frac{1}{T} - \frac{1}{T_{\mathrm{ref}}}\right)\right], \tag{12}$$

where $E_{a,Q} = 3.2$ kJ·mol^{-1} and Q_{init} is calibrated per chemistry: for Samsung INR18650-20R, nameplate rating is $Q_{\mathrm{nom}} = 2.00$ A·h ($Q_{\mathrm{init}} = 2.00$ A · h); for A123 ANR26650M1-B, datasheet nameplate is $Q_{\mathrm{nom}} = 2.50$ A · h (2.40 A · h min), whereas measured dynamic usable capacity is $Q_{\mathrm{init}} = 2.30$ A · h (2.00–3.60 V at 1C, 25°C). Across aging (SOH = 1.0 → 0.80), series resistance $R_0 = R_{\mathrm{ohm}} + R_{\mathrm{sei}}$ expands by 50% (35.0 to 52.5 mΩ) and charge-transfer resistance R_1 expands by 150% (15.0 to 37.5 mΩ). Evaluating $\tilde{D}_{U_i} = R_i(T, \mathrm{SOH})I_{\mathrm{bias}} + \delta R_i I_{\mathrm{RMS}}$ with current sensor accuracy $I_{\mathrm{bias}} \leq 0.005$ A (Arbin ±0.05% F.S.), $\delta R_i \leq 0.0028\,\Omega$, and drive-cycle $I_{\mathrm{RMS}} = 1.0$ A yields $\tilde{D}_{U_i} \leq 3.5$ mV across 0°C–45°C (peaking at 0°C with $R_1 \approx 107$ mΩ, and ≤ 2.2 mV at 25°C). Crucially, we enforce a strict parameter allocation between plant and estimators: in the battery plant (Table 10), aging modulates $R_0, R_1, R_2, C_1, C_2, Q_{\mathrm{act}}$; uncompensated observers (EKF, AEKF, SMO) freeze all parameters at fresh nominal values to quantify degradation drift; standard DEKF adapts $\hat{\boldsymbol{\theta}}_k = [R_{0,k}, 1/Q_{\mathrm{act},k}]^T$ while purposefully freezing R_1, R_2, C_1, C_2 to prevent structural parameter collinearity and unidentifiability; unmodeled polarization residual $\delta V^*_{\mathrm{pol},k} \equiv (U^{\mathrm{plant}}_{1,k} - \hat{U}^{\mathrm{ECM}}_{1,k}) + (U^{\mathrm{plant}}_{2,k} - \hat{U}^{\mathrm{ECM}}_{2,k})$ is compensated online by the proposed PIDKF.

2.4 OCV-SOC Monotonicity and Dynamic Hysteresis. To prevent artificial divergence in gradient-based observers (Runge's phenomenon), the $V_{\mathrm{ocv}}(\mathrm{SOC})$ function for NMC chemistry is strictly calibrated as a monotonically increasing 5th-order polynomial satisfying $dV_{\mathrm{ocv}}/d\mathrm{SOC} > 0$ for all SOC $\in [0, 1]$:

$$V_{\mathrm{ocv}}(z) = p_5z^5 + p_4z^4 + p_3z^3 + p_2z^2 + p_1z + p_0, \tag{13}$$

where $z \equiv$ SOC, with calibrated polynomial coefficients: $p_5 = 9.204$, $p_4 = -28.841$, $p_3 = 33.376$, $p_2 = -17.419$, $p_1 = 4.674$, and $p_0 = 3.212$. Differentiating yields the exact quartic derivative: $V'_{\mathrm{ocv}}(z) = 46.020z^4 - 115.364z^3 + 100.128z^2 - 34.838z + 4.674$. All four roots of $V'_{\mathrm{ocv}}(z) = 0$ are complex conjugate pairs ($z_{1,2} \approx 0.964 \pm 0.156j$, $z_{3,4} \approx 0.289 \pm 0.151j$), guaranteeing $V'_{\mathrm{ocv}}(z) > 0$ strictly for all $z \in \mathbb{R}$. Over the physical domain $z \in [0, 1]$, $V'_{\mathrm{ocv}}(z) \in [\alpha, V'_{\max}] = [0.481, 4.674]$ V, bounded away from zero. By the Inverse Function Theorem, $z = V^{-1}_{\mathrm{ocv}}(V)$ is uniquely defined and regular with $(V^{-1}_{\mathrm{ocv}})'(V) \in [0.214, 2.079]$ V^{-1}.

For the three-state NMC equivalent circuit, the discrete three-step observability matrix O_k possesses a structured Vandermonde determinant:

$$\det(O_k) = V'_{\mathrm{ocv}}(z_k)(1 - a_1)(1 - a_2)(a_1 - a_2). \tag{14}$$

Since $\tau_1 = 24.0$ s $\neq \tau_2 = 112.5$ s ($a_1 \neq a_2$), $\Delta t > 0$, and $V'_{\mathrm{ocv}}(z) \geq \alpha > 0$, we have $\det(O_k) \neq 0$ and $\mathrm{rank}(O_k) = 3$ everywhere on $z \in [0, 1]$, guaranteeing uniform local observability. Conversely, when $V'_{\mathrm{ocv}}(z) \to 0$ (as in the LFP flat plateau where $V'_{\mathrm{ocv}} \approx 0.08$ V), $\det(O_k)$ shrinks by an order of magnitude and condition number $\kappa(O_k)$ surges from 3.2×10^4 to $> 1.99 \times 10^5$, establishing the analytical root cause of flat-plateau observability degradation.

For $LiFePO_4$ (LFP) chemistry, open-circuit voltage exhibits a wide, flat biphasic transition plateau ($LiFePO_4 \leftrightarrow FePO_4$) where $dV_{ocv}/dz \approx 0.08$ V between 20% and 80% SOC, accompanied by non-negligible dynamic hysteresis [13,14]. Governed by the Gibbs phase rule ($F = C - P + 2$; at fixed temperature and pressure, degrees of freedom become $F' = C - P = 2 - 2 = 0$), two-phase coexistence pins the lithium chemical potential to an invariant value independent of stoichiometry, producing the near-zero voltage slope and severe observability loss. Across the operational domain ($z \in [0.12, 0.95]$), LFP open-circuit voltage is parameterized via $V_{ocv}^{LFP}(z) = 3.415 - 0.286z + 1.831z^2 - 4.120z^3 + 4.185z^4 - 1.545z^5$ (RMS error 2.4 mV). Direct evaluation of $V'_{ocv}(z) = -0.286 + 3.662z - 12.36z^2 + 16.74z^3 - 7.725z^4$ confirms strict monotonicity $V'_{ocv}(z) \geq \alpha_{LFP} = 0.00275$ V > 0 for all $z \in [0.12, 0.95]$ ($V'_{ocv}(0.12) = 0.00278$ V, rising to > 0.065 V over the central 20%–80% plateau, with a stationary point (local minimum of V_{ocv}) at $z \approx 0.118$; deep discharge $z < 0.12$ is prevented by BMS lower voltage cutoff protection ($V_t \leq 2.00$ V) [15]). Hysteresis parameters are $\beta = 45.0$ and $M(z) \equiv M_0 = 30$ mV ($M_{max} = 35$ mV):

$$\frac{dV_h(t)}{dt} = -\frac{\beta\eta|I(t)|}{3600\,Q_{act}(T,\mathrm{SOH})}\left[V_h(t) - M(z)\,\mathrm{sgn}(I(t))\right], \quad (15)$$

where $V_h(t)$ is hysteresis voltage, β is decay rate, and $M(z)$ is maximum hysteresis voltage ($M \approx$ 20–40 mV). Evaluating Eq. (15) across interval $[t_k, t_{k+1})$ under zero-order hold (ZOH) current I_k yields the exact discrete-time analytical solution:

$$V_{h,k+1} = e^{-\beta|\Delta z_k|}V_{h,k} + \left(1 - e^{-\beta|\Delta z_k|}\right)M(z_k)\,\mathrm{sgn}(I_k), \quad (16)$$

with $|\Delta z_k| = \frac{\eta|I_k|\Delta t}{3600 Q_{act}}$. Because $e^{-\beta|\Delta z_k|} \in (0, 1]$, Eq. (16) represents a convex combination (contractive with $a_h < 1$ for $I_k \neq 0$; static with $a_h = 1$ at rest $I_k = 0$) bounded by $|V_{h,k+1}| \leq \max(|V_{h,k}|, M(z_k)) \leq M_{max}$, proving $[-M_{max}, M_{max}]$ is positively invariant.

For LFP chemistry, the state vector is augmented with dynamic hysteresis: $\mathbf{x}_k^{LFP} = [\mathrm{SOC}_k, U_{1,k}, U_{2,k}, V_{h,k}]^T$. The discrete evolution follows $\mathbf{x}_{k+1}^{LFP} = \mathbf{A}_k^{LFP}\mathbf{x}_k^{LFP} + \mathbf{B}_k^{LFP}I_k + \mathbf{g}_k + \mathbf{w}_k$ with $\mathbf{A}_k^{LFP} = \mathrm{diag}(1, e^{-\Delta t/\tau_1}, e^{-\Delta t/\tau_2}, e^{-\beta|\Delta z_k|})$ (since constant plateau saturation $M(z) \equiv M_0$ yields $M'(z) \equiv 0$ and decoupled hysteresis in $\mathbf{A}_k^{LFP}$), $\mathbf{B}_k^{LFP} = [B_{1,k}, B_{2,k}, B_{3,k}, 0]^T$, and nonlinear excitation $\mathbf{g}_k = [0, 0, 0, (1 - e^{-\beta|\Delta z_k|})M(z_k)\,\mathrm{sgn}(I_k)]^T$. The terminal voltage satisfies:

$$V_{t,k} = V_{ocv}(\mathrm{SOC}_k) - V_{h,k} - U_{1,k} - U_{2,k} - I_k R_0 + v_k. \quad (17)$$

For the augmented four-state LFP formulation, incorporating the dynamic hysteresis mode expands the discrete observability determinant into:

$$|\det(\mathcal{O}_{4,k})| = V'_{ocv}(z_k)(1 - a_h)|a_1 - a_2|\prod_{i=1}^{2}(1 - a_i)|a_i - a_h|, \quad (18)$$

where $a_{h,k} = e^{-\beta|\Delta z_k|}$. Equation (18) is a frozen-coefficient local rank test over short windows ($a_{h,k} \approx a_h$). Under rest ($I_k = 0$), $|\Delta z_k| = 0 \implies a_{h,k} = 1 = \lambda_{SOC}$, so $1 - a_{h,k} = 0$ and $\mathrm{rank}(\mathcal{O}_{4,k}) \leq 3$: static hysteresis is collinear with V_{ocv} and unobservable from voltage alone. Conversely, for the linear time-varying (LTV) system under *Persistent Excitation* (PE, $\exists \delta_I > 0, T_{PE} \geq 1$ s.t. $\frac{1}{T_{PE}}\sum_{j=k}^{k+T_{PE}-1}|I_j| \geq \delta_I > 0$), $a_{h,k} < 1$ separates from unity and (a_1, a_2). With discrete pole $a_{h,k} \in (0, 1)$, the non-resonance condition $a_{h,k} \neq a_1, a_2$ ensures distinct eigenvalues. Over active PE windows of horizon $N = 30$ steps ($\Delta t = 1.0$ s) under $a_{h,k} \neq a_1, a_2$, the discrete LTV Gramian $\mathcal{G}_o(k, k+N) = \sum_{m=k}^{k+N-1}\mathbf{\Phi}_A^T(m,k)\tilde{\mathbf{H}}_m^T\tilde{\mathbf{H}}_m\mathbf{\Phi}_A(m,k) \succeq \alpha_o\mathbf{I}$ ($\mathbf{\Phi}_A(m,k) = \mathbf{A}_{m-1}\cdots\mathbf{A}_k$) on normalized states $\tilde{\mathbf{x}} = [\mathrm{SOC}, U_1/V_{nom}, U_2/V_{nom}, V_h/M_{max}]^T$ ($V_{nom} = 3.30$ V, $M_{max} = 0.035$ V) yields $\lambda_{min}(\mathcal{G}_o) \geq 1.42 \times 10^{-3} > 0$, while flat-plateau $\kappa(\mathcal{O}_{4,k})$ surges from 3.2×10^4 to $> 1.99 \times 10^5$ as $V'_{ocv} \to 0$. During rest ($I_k = 0$), parameter updates are frozen ($\mathbf{K}_{\theta,k} = \mathbf{0}$).

3 State-of-Charge Estimation Methodologies

3.1 Extended Kalman Filter (EKF). The benchmark investigates EKF, AEKF, DEKF, SMO, BiLSTM-Attn, and PINN-SOC as archetype representatives of classical model-based, adaptive, nonlinear sliding-mode, and physics-informed machine learning paradigms. Higher-order sigma-point filters such as UKF and CKF [9] are omitted from the primary benchmark because the 2nd-order Taylor expansion error of the 5th-order OCV polynomial is < 0.015 mV (inducing < 0.02% SOC deviation); EKF already achieves RMSE = 0.065% under nominal conditions (approaching the ±0.036% reference noise floor). Because UKF and CKF share the identical fixed-parameter structure as EKF, they suffer the exact same catastrophic 45-fold accuracy collapse under battery aging while imposing a $5\times$–$7\times$ computational penalty ($2n + 1$ sigma points). The EKF linearizes the nonlinear measurement equation at each step around prior estimate $\hat{\mathbf{x}}_k^-$ via the Jacobian vector $\mathbf{H}_k$ [16]:

$$\mathbf{H}_k^{NMC} = \begin{bmatrix} V'_{ocv}(\hat{z}_k^-) & -1 & -1 \end{bmatrix},$$

$$\mathbf{H}_k^{LFP} = \begin{bmatrix} V'_{ocv}(\hat{z}_k^-) & -1 & -1 & -1 \end{bmatrix}, \quad (19)$$

where $\partial V_t/\partial V_h = -1$ reflects the subtractive nature of discharge hysteresis overpotential. The recursive algorithm executes in two alternating stages:

1. Time Update (Predict):

$$\hat{\mathbf{x}}_k^- = \mathbf{A}_{k-1}\hat{\mathbf{x}}_{k-1}^+ + \mathbf{B}_{k-1}I_{k-1} + \mathbf{g}_{k-1} \quad (20)$$

$$\mathbf{P}_k^- = \mathbf{A}_{k-1}\mathbf{P}_{k-1}^+\mathbf{A}_{k-1}^T + \mathbf{Q} \quad (21)$$

where $\mathbf{g}_{k-1} = \mathbf{0}$ for NMC and satisfies Eq. (16) for LFP.

2. Measurement Update (Correct):

$$e_k = V_k - \hat{V}_{t,k}^-, \quad \mathbf{K}_k = \mathbf{P}_k^-\mathbf{H}_k^T\left(\mathbf{H}_k\mathbf{P}_k^-\mathbf{H}_k^T + R_v\right)^{-1} \quad (22)$$

$$\hat{\mathbf{x}}_k^+ = \hat{\mathbf{x}}_k^- + \mathbf{K}_k e_k \quad (23)$$

$$\mathbf{P}_k^+ = (\mathbf{I} - \mathbf{K}_k\mathbf{H}_k)\mathbf{P}_k^-(\mathbf{I} - \mathbf{K}_k\mathbf{H}_k)^T + \mathbf{K}_k R_v \mathbf{K}_k^T \quad (24)$$

with $\hat{V}_{t,k}^- = V_{ocv}(\hat{z}_k^-) - \hat{U}_{1,k}^- - \hat{U}_{2,k}^- - I_k R_0$ (augmented with $-\hat{V}_{h,k}^-$ for LFP). Equation (24) implements the stabilized Joseph covariance form, which guarantees symmetry ($\mathbf{P}_k^+ = (\mathbf{P}_k^+)^T$) and strict positive-definiteness ($\mathbf{P}_k^+ \succ 0$) under finite-precision floating-point arithmetic.

3.2 Adaptive Extended Kalman Filter (AEKF). Fixed noise covariances ($\mathbf{Q}, R_v$) degrade filtering performance when measurement conditions vary. The AEKF integrates the Sage-Husa online adaptive noise estimator [18]:

$$\hat{r}_k = (1 - d_k)\hat{r}_{k-1} + d_k\left(V_k - \hat{V}_k^-\right) \quad (25)$$

$$\hat{R}_{v,k} = (1 - d_k)\hat{R}_{v,k-1} + d_k\left(e_k^2 - \mathbf{H}_k\mathbf{P}_k^-\mathbf{H}_k^T\right) \quad (26)$$

where $d_k = (1 - \lambda_{SH})/(1 - \lambda_{SH}^k)$ is an exponential weighting factor with forgetting factor $\lambda_{SH} = 0.985$. To prevent filter divergence arising from the structural unidentifiability of simultaneously estimating process noise $\mathbf{Q}_k$ and measurement noise R_k from a single scalar innovation sequence (classic Sage-Husa theorem), $\mathbf{Q}$ is fixed to its calibrated baseline while $\hat{R}_{v,k}$ is updated with a semi-definite projection constraint $\hat{R}_{v,k} = \max(R_{v,min}, \hat{R}_{v,k})$ to guarantee positive-definiteness.

3.3 Dual Extended Kalman Filter (DEKF). To decouple fast state estimation from slow parameter degradation, the DEKF employs two interacting Kalman filters running in parallel [16,23]: (1) *State Filter*: Estimates $\mathbf{x}_k = [\mathrm{SOC}_k, U_{1,k}, U_{2,k}]^T$ using current parameter estimates $\hat{\boldsymbol{\theta}}_k = [\hat{R}_{0,k}, 1/\hat{Q}_{\mathrm{act},k}]^T$; (2) *Parameter Filter*: Models parameter evolution as a random walk $\boldsymbol{\theta}_{k+1} = \boldsymbol{\theta}_k + \mathbf{w}_{\theta,k}$ with $\mathbf{w}_{\theta,k} \sim \mathcal{N}(0, \mathbf{Q}_\theta)$. The parameter sensitivity matrix $\mathbf{C}_{\theta,k} = \left.\frac{dV_k}{d\theta}\right|_{\hat{\theta}_k^-}$ is derived via total Fréchet differentiation:

$$\mathbf{C}_{\theta,k} = \left[-I_k \quad -\tfrac{\eta \Delta t I_{k-1}}{3600} V'_{\mathrm{ocv}}(\hat{z}_k^+)\right], \tag{27}$$

where $\frac{\partial V_k}{\partial \theta_2} = \frac{\partial V_{\mathrm{ocv}}}{\partial \mathrm{SOC}_k} \frac{\partial \mathrm{SOC}_k}{\partial (1/Q_{\mathrm{act}})} = V'_{\mathrm{ocv}}\left(-\frac{\eta \Delta t I_{k-1}}{3600}\right)$. The parameter filter updates recursively using the stabilized Joseph covariance form:

$$\mathbf{K}_{\theta,k} = \mathbf{P}^-_{\theta,k} \mathbf{C}^T_{\theta,k} [\mathbf{C}_{\theta,k} \mathbf{P}^-_{\theta,k} \mathbf{C}^T_{\theta,k} + R_{\theta,v}]^{-1}, \tag{28}$$

$$\hat{\boldsymbol{\theta}}_k^+ = \hat{\boldsymbol{\theta}}_k^- + \mathbf{K}_{\theta,k} e_k, \tag{29}$$

$$\mathbf{P}^+_{\theta,k} = (\mathbf{I} - \mathbf{K}_{\theta,k}\mathbf{C}_{\theta,k})\mathbf{P}^-_{\theta,k}(\mathbf{I} - \mathbf{K}_{\theta,k}\mathbf{C}_{\theta,k})^T + \mathbf{K}_{\theta,k} R_{\theta,v} \mathbf{K}^T_{\theta,k} \tag{30}$$

3.4 Sliding Mode Observer (SMO). The SMO provides robust tracking against bounded modeling disturbances by driving the voltage error onto a boundary layer around sliding manifold $S_k = V_{\mathrm{meas},k} - \hat{V}_{t,k} = 0$ [25,26]:

$$\hat{\mathbf{x}}_{k+1} = \mathbf{A}_k \hat{\mathbf{x}}_k + \mathbf{B}_k I_k + \mathbf{g}_k + \mathbf{L}(\hat{z}_k)\,\mathrm{sat}\left(\frac{S_k}{\mu}\right), \tag{31}$$

where $\mu = 0.02$ V is boundary thickness and continuous saturation is $\mathrm{sat}(S/\mu) = \max(-1, \min(1, S/\mu))$. To eliminate gain explosion and maintain reachability across both steep NMC slopes and flat LFP plateaus, feedback injection is scheduled onto the SOC coordinate: $\mathbf{L}(\hat{z}) = [l_{\mathrm{smo}}/\max(\alpha_{\min}, V'_{\mathrm{ocv}}(\hat{z})), 0, 0, (0)]^T$, with scalar gain $l_{\mathrm{smo}} = 0.025$ V and $\alpha_{\min} = 0.40/0.00275$ V for NMC/LFP. Because $\mathbf{H}_k = [V'_{\mathrm{ocv}}, -1, -1, (-1)]$, the effective manifold injection gain is state-invariant: $\gamma(\hat{z}) \equiv \mathbf{H}_k \mathbf{L}(\hat{z}) = l_{\mathrm{smo}} = 0.025\ \mathrm{V} > \Phi_d$ whenever $V'_{\mathrm{ocv}} \geq \alpha_{\min}$. Polarization sub-states (U_1, U_2) and hysteresis (V_h) evolve open-loop along passive dissipative dynamics $(l_2 = l_3 = l_h = 0)$, preventing chattering and ensuring discrete quasi-sliding mode stability as proven in Appendix B. All benchmark metrics and Table 8 timings (281 cycles on Cortex-M4, including polynomial OCV derivative and gain scheduling) correspond directly to this implementation.

3.5 Bidirectional LSTM with Self-Attention (BiLSTM-Attn). The deep data-driven architecture processes a temporal sliding window $\mathbf{X}_k = [I_{k-W:k}, V_{k-W:k}, T_{k-W:k}] \in \mathbb{R}^{W\times 3}$ with window length $W = 30$ s [35,37]. Forward $(\overrightarrow{\mathbf{h}}_t)$ and backward $(\overleftarrow{\mathbf{h}}_t)$ hidden states are concatenated: $\mathbf{h}_t = [\overrightarrow{\mathbf{h}}_t \,\|\, \overleftarrow{\mathbf{h}}_t] \in \mathbb{R}^{2D}$. The temporal self-attention mechanism computes alignment scores:

$$u_t = \mathbf{v}^T \tanh(\mathbf{W}_a \mathbf{h}_t + \mathbf{b}_a), \quad \alpha_t = \frac{\exp(u_t)}{\sum_{j=1}^{W} \exp(u_j)} \tag{32}$$

$$\mathbf{c}_k = \sum_{t=1}^{W} \alpha_t \mathbf{h}_t \tag{33}$$

where $\mathbf{c}_k$ represents the context vector fed into fully connected layers to predict $\hat{z}_k$.

3.6 Physics-Informed Neural Network (PINN-SOC). Unlike conventional black-box networks that treat SOC estimation as an unconstrained regression task, the Physics-Informed Neural Network (PINN-SOC) embeds first-principles electrochemical conservation laws directly into the backpropagation graph [39–41].

1. Continuous State-Space and Automatic Differentiation: From macroscopic Faradaic charge conservation, battery charge depletion satisfies the continuous governing differential equation:

$$\frac{d\,\mathrm{SOC}(t)}{dt} = -\frac{\eta(T, I)\, I(t)}{3600\, Q_{\mathrm{act}}(T, \mathrm{SOH})}, \tag{34}$$

where $\eta \in (0, 1]$ represents Faradaic coulombic efficiency capturing side reactions, and Q_{act} denotes usable capacity modulated by Arrhenius thermal kinetics and SEI growth. A deep surrogate $\mathcal{NN}(\mathbf{u}(t); \boldsymbol{\theta})$ receives input vector $\mathbf{u}(t) = [t, I, V_t, T, \mathrm{SOH}]^T$ and predicts $[\hat{z}(t), \hat{V}_{\mathrm{pol}}(t)]^T$. Because cell capacity degradation occurs on macroscopic timescales (10^5–10^6 s), across any individual drive schedule (600–3600 s) the time derivative of SOH is negligible: $d\mathrm{SOH}/dt \approx 0$ ($|d\mathrm{SOH}/dt| < 10^{-8}$ s^{-1}). Hence, the total time derivative $\frac{d\hat{z}}{dt}$ along driving trajectories is evaluated via chain-rule Autograd:

$$\frac{d\hat{z}(t)}{dt} = \frac{\partial \mathcal{NN}_1}{\partial t} + \frac{\partial \mathcal{NN}_1}{\partial I}\frac{dI}{dt} + \frac{\partial \mathcal{NN}_1}{\partial V_t}\frac{dV_t}{dt} + \frac{\partial \mathcal{NN}_1}{\partial T}\frac{dT}{dt}, \tag{35}$$

$$\mathcal{R}_{\mathrm{Faraday}}(t) = \frac{d\hat{z}(t)}{dt} + \frac{\eta(T, I)\, I(t)}{3600\, Q_{\mathrm{act}}(T, \mathrm{SOH})}, \tag{36}$$

where telemetry derivatives $(dI/dt, dV_t/dt, dT/dt)$ are computed offline via smoothing splines during training loss backpropagation, while online BMS inference executes strictly causally.

2. Electrochemical KVL Consistency and Thermodynamic Bounds: Operational battery pack sensors cannot non-invasively measure internal overpotentials (U_1, U_2). PINN-SOC resolves this observability deficit by enforcing Kirchhoff's Voltage Law (KVL) as an electrochemical consistency manifold (where $\hat{V}_{\mathrm{pol}}(t) \equiv \hat{U}_1 + \hat{U}_2 + \hat{V}_h$ lumps dynamic polarization and hysteresis):

$$\mathcal{R}_{\mathrm{KVL}}(t) = V_{\mathrm{meas}}(t) - \left[V_{\mathrm{ocv}}(\hat{z}(t)) - I(t)R_0(T, \mathrm{SOH}) - \hat{V}_{\mathrm{pol}}(t)\right]. \tag{37}$$

Thermodynamic equilibrium stability is anchored by the strict monotonicity of the open-circuit voltage curve $(dV_{\mathrm{ocv}}/d\,\mathrm{SOC} > 0)$, while an output Sigmoid activation intrinsically guarantees physical boundary invariance $\hat{z} \in (0, 1)$.

3. Composite Multi-Objective Loss Formulation: The training loss balances supervised fidelity over labeled tuples $\mathcal{D}_{\mathrm{data}} = \{(t_k, I_k, V_{t,k}, T_k, \mathrm{SOH}_k, \mathrm{SOC}_k^{\mathrm{ref}})\}_{k=1}^{N_d}$ (with normalized elapsed cycle time $t_k = k\Delta t$ and RPT-calibrated SOH_k) against physics residuals over collocation points $\mathcal{D}_{\mathrm{colloc}} = \{(t_j, I_j, V_{t,j}, T_j, \mathrm{SOH}_j)\}_{j=1}^{N_c}$:

$$\mathcal{L}_{\mathrm{total}}(\boldsymbol{\theta}) = \mathcal{L}_{\mathrm{data}} + \lambda_{\mathrm{Faraday}}\mathcal{L}_{\mathrm{Faraday}} + \lambda_{\mathrm{KVL}}\mathcal{L}_{\mathrm{KVL}} + \lambda_{\mathrm{mono}}\mathcal{L}_{\mathrm{mono}}, \tag{38}$$

where $\mathcal{L}_{\mathrm{data}} = \frac{1}{N_d}\sum_{k=1}^{N_d}(\hat{z}_k - \mathrm{SOC}_k^{\mathrm{ref}})^2$ denotes supervised error, $\mathcal{L}_{\mathrm{Faraday}} = \frac{1}{N_c}\sum_{j=1}^{N_c}\|\mathcal{R}_{\mathrm{Faraday}}(t_j)\|^2$ and $\mathcal{L}_{\mathrm{KVL}} = \frac{1}{N_c}\sum_{j=1}^{N_c}\|\mathcal{R}_{\mathrm{KVL}}(t_j)\|^2$ enforce physical conservation, and $\mathcal{L}_{\mathrm{mono}} = \frac{1}{N_c}\sum_{j=1}^{N_c}\max(0, -\partial\hat{z}_j/\partial V_{t,j})^2$ acts as a quasi-stationary network monotonicity regularization ($\partial\hat{z}/\partial V_t \geq 0$ at fixed (I, T, SOH) to suppress non-monotonic artifacts, distinct from equilibrium $dV_{\mathrm{ocv}}/d\,\mathrm{SOC} > 0$; $\lambda_{\mathrm{Faraday}} = 0.40$, $\lambda_{\mathrm{KVL}} = 0.30$, $\lambda_{\mathrm{mono}} = 0.10$). The 4-layer perceptron (input dim 5, hidden layers $[64, 64, 32]$ with GELU activations, output dim 2) is trained via AdamW ($\eta_0 = 10^{-3}$ to 10^{-5}, 500 epochs, batch size 256) across 5 random seeds.

3.7 Proposed Hybrid Architecture: PIDKF. To combine closed-loop Lyapunov stability and online parameter tracking with nonlinear neural representation, we formulate the hybrid **Physics-Informed Dual Kalman Filter (PIDKF)**. A dual extended Kalman filter co-estimates state $\hat{\mathbf{x}}_k$ (3-state for NMC; 4-state including hysteresis $\hat{V}_{h,k}$ via Eq. (17) for LFP) and parameters $\hat{\boldsymbol{\theta}}_k = [R_{0,k}, 1/Q_{\text{act},k}]^T$, while a compact neural compensator with input $\mathbf{u}_k = [I_k, T_k, \hat{z}_k^-, \hat{U}_{1,k}^-, \hat{U}_{2,k}^-, \text{SOH}_{\text{ref},k}]^T \in \mathbb{R}^6$ (where reference degradation SOH at $T_{\text{ref}} = 25^\circ$C inverts thermal factor: $\text{SOH}_{\text{ref},k} \equiv \frac{\hat{Q}_{\text{act},k}(T_k)}{Q_{\text{init}} f_Q(T_k)} = \frac{1}{Q_{\text{init}} \theta_{2,k} f_Q(T_k)}$) predicts unmodeled kinetic overpotential $\delta\hat{V}_{\text{pol},k} = \text{clip}(\mathcal{NN}(\mathbf{u}_k; \boldsymbol{\theta}_{\text{net}}), -V_{\text{pol}}^{\max}(I_k), V_{\text{pol}}^{\max}(I_k))$, enforcing $V_{\text{pol}}^{\max}(I_k) = \kappa_0 + \kappa_1 |I_k| \le \kappa_0 + \kappa_1 I_{\max}$. Offline training on Cell Batch A uses reference $\text{SOC}_k^{\text{ref}}$ to form target $\delta V_{\text{pol},k}^{\text{target}} \equiv V_{\text{meas},k} - [V_{\text{ocv}}(\text{SOC}_k^{\text{ref}}) - V_{h,k} - U_{1,k}^{\text{ECM}} - U_{2,k}^{\text{ECM}} - I_k R_0(\text{SOH})]$, isolating polarization mismatch from state drift. Online, $\hat{V}_{t,\text{PIDKF},k}^- = \hat{V}_{t,\text{ECM},k}^- + \delta\hat{V}_{\text{pol},k}$ yields innovation $e_{k,\text{PIDKF}} = V_{\text{meas},k} - \hat{V}_{t,\text{PIDKF},k}^-$, driving $\hat{\mathbf{x}}_k^+ = \hat{\mathbf{x}}_k^- + \mathbf{K}_k e_{k,\text{PIDKF}}$ and $\hat{\boldsymbol{\theta}}_k^+ = \hat{\boldsymbol{\theta}}_k^- + \mathbf{K}_{\theta,k} e_{k,\text{PIDKF}}$ with effective innovation covariance $R_{\text{eff},k} = R_v + \sigma_{\text{NN}}^2 + \sigma_{\text{approx}}^2 = 1.04\times10^{-3}$ V^2 for the state filter (where zero-mean orthogonality $\mathbb{E}[e_{\text{NN}} e_{\text{approx}}] = 0$ between residual $\sigma_{\text{NN}} = 5.8$ mV and truncation $\sigma_{\text{approx}} = 1.9$ mV justifies additive variance expansion; parameter covariance similarly expands via $R_{\theta,\text{eff}} = R_{\theta,v} + \sigma_{\text{NN}}^2$). Exact differentiation w.r.t. $\hat{\mathbf{x}}_k^-$ and $\hat{\boldsymbol{\theta}}_k = [\hat{R}_{0,k}, 1/\hat{Q}_{\text{act},k}]^T$ (with $\theta_{2,k} = 1/\hat{Q}_{\text{act},k}$) yields $\mathbf{H}_{\text{PIDKF},k}^{\text{exact}} = \mathbf{H}_k + \partial\delta\hat{V}_{\text{pol},k}/\partial\hat{\mathbf{x}}_k^-$ and exact parameter sensitivity $\mathbf{C}_{\theta,\text{PIDKF},k}^{\text{exact}} = \mathbf{C}_{\theta,k} + [0,\ \frac{\partial\mathcal{NN}}{\partial\text{SOH}_{\text{ref},k}}(-\frac{\hat{Q}_{\text{act},k}^2}{Q_{\text{init}} f_Q(T_k)})]$. In embedded PIDKF, we implement an *exogenous sensitivity decoupling* ($\mathbf{H}_{\text{PIDKF},k} \approx \mathbf{H}_k$, $\mathbf{C}_{\theta,\text{PIDKF},k} \approx \mathbf{C}_{\theta,k}$) by treating SOH as quasi-static ($\frac{\partial\mathcal{NN}}{\partial\text{SOH}_{\text{ref},k}} \approx 0$ during instantaneous parameter steps) to omit neural backpropagation. In normalized coordinates $\tilde{\mathbf{x}}_k^{\text{NMC}}$ and $\tilde{\mathbf{x}}_k^{\text{LFP}}$, $\max_k \|\tilde{\mathbf{H}}_{\text{PIDKF},k} - \tilde{\mathbf{H}}_k\|_2 = 0.0184$ V (NMC, $< 3.8\%$) and 0.0162 V (LFP, $< 3.4\%$). The exogenous sensitivity decoupling is quantitatively justified by the small parameter sensitivity perturbation $\max_k \|\mathbf{C}_{\theta,\text{PIDKF},k}^{\text{exact}} - \mathbf{C}_{\theta,k}\|_2 \le 0.0041$ V $\cdot$ A $\cdot$ h (inducing $< 0.12\%$ gain discrepancy $\Delta\mathbf{K}_\theta$); parameter errors between Exact and Exogenous are virtually identical (R_0: 1.21% vs. 1.23%; Q_{act}: 0.82% vs. 0.84%; Table 9), saving 1,420 cycles/step (37.3%) on Cortex-M4.

The neural compensator is optimized via the composite loss:

$$\mathcal{L}_{\text{PIDKF}}(\boldsymbol{\theta}_{\text{net}}) = \mathcal{L}_{\text{res}} + \lambda_{\text{phys}}\mathcal{L}_{\text{phys}} + \lambda_{\text{reg}}\|\boldsymbol{\theta}_{\text{net}}\|_2^2, \tag{39}$$

where $\mathcal{L}_{\text{res}} = \frac{1}{N}\sum_{k=1}^N (\delta V_{\text{pol},k}^{\text{target}} - \delta\hat{V}_{\text{pol},k})^2$, and the physics constraint enforces a linearized Tafel/ohmic overpotential envelope $V_{\text{pol}}^{\max}(I_k) = \kappa_0 + \kappa_1|I_k|$ and dissipative polarity ($I_k\delta\hat{V}_{\text{pol},k} \le 0$): $\mathcal{L}_{\text{phys}} = \frac{1}{N}\sum_{k=1}^N [\max(0, |\delta\hat{V}_{\text{pol},k}| - V_{\text{pol}}^{\max})^2 + \gamma_I \max(0, I_k\delta\hat{V}_{\text{pol},k})^2]$, preventing the compensator from acting as an unphysical energy source or fitting noise. The neural compensator uses a 2-hidden-layer MLP ([32, 32] neurons, GELU activations, 1,313 parameters = 5.25 KB Flash/SRAM, AdamW with learning rate 10^{-3}, weight decay 10^{-4}, 200 epochs, batch size 128).

Stability Assumptions and Error Boundedness: We formalize PIDKF boundedness under two assumptions: *Assumption 1 (Hard Overpotential Bound)*: True overpotentials satisfy $|\delta V_{\text{pol},k}^*| \le V_{\text{pol,max}} < \infty$. Via clipping, $|\delta\hat{V}_{\text{pol},k}| \le \kappa_0 + \kappa_1 I_{\max}$ holds identically, bounding total innovation mismatch $\sup_k |\varepsilon_k| \le \bar{\varepsilon} < \infty$ (empirically $\sup_k |\varepsilon_k| = 14.2$ mV on aged NMC and 9.6 mV on aged LFP hold-out sets, defining the validated empirical residual envelope $\bar{\varepsilon} = 20.0$ mV across the tested domain). *Assumption 2 (Intermittent Observability and Bounded Rest)*: Active PE windows $[k_m, k_m + N]$ ($N = 30$) satisfying Uniform Complete Observability (UCO, $\mathcal{G}_o \succeq \alpha_o \mathbf{I}$) and Controllability (UCC) are separated by bounded rest intervals $T_{\text{rest},m} \le T_{\max} = 120$ s ($I_k = 0$). During rest, parameter adaptation is frozen ($\mathbf{K}_{\theta,k} = \mathbf{0}$) and closed-loop state dynamics satisfy non-expansiveness $\|\mathbf{F}_k\|_2 \le 1$ (enforced via hard rest-period innovation gating: setting $\mathbf{K}_k = \mathbf{0}$ and $\mathbf{K}_{\theta,k} = \mathbf{0}$ whenever $|I_k| < I_{\text{rest,th}} = 10$ mA, identically yielding $\mathbf{F}_k = \mathbf{A}_k$ and $\|\mathbf{F}_k\|_2 = \|\mathbf{A}_k\|_2 \le 1$, or via bounded Kalman gains $\|(\mathbf{I} - \mathbf{K}_k\mathbf{H}_k)\|_2 \le 1$), with $\underline{q}\mathbf{I} \preceq \mathbf{Q}_k \preceq \bar{q}\mathbf{I}$ and $0 < \underline{r} \le R_{\text{eff},k} \le \bar{r} < \infty$.

Under Assumptions 1–2, each cycle of length $L_m = N + T_{\text{rest},m} \le N + T_{\max}$ contracts by $\|\boldsymbol{\Phi}_F(k_m + L_m, k_m)\| \le \rho_{\text{PE}} < 1$. Partitioning any arbitrary interval $[j, k]$ into $M = \lfloor (k-j)/(N+T_{\max}) \rfloor$ complete cycles and boundary partial windows of length $\le N + T_{\max}$ bounded by $\bar{F}^{N+T_{\max}}$ ($\bar{F} \ge \sup_k \|\mathbf{F}_k\|$), we define composite decay $\rho \equiv \rho_{\text{PE}}^{1/(N+T_{\max})} < 1$ and lead prefactor $c \equiv \bar{F}^{N+T_{\max}} \rho^{-(N+T_{\max})} \ge 1$, ensuring uniform exponential decay $\|\boldsymbol{\Phi}_F(k, j)\| \le c\rho^{k-j}$ for all $k \ge j$, bounded covariance $\underline{p}\mathbf{I} \preceq \mathbf{P}_k \preceq \bar{p}\mathbf{I}$, and gain $\|\mathbf{K}_k\| \le \bar{K} \equiv \bar{p}\bar{H}/\underline{r} < \infty$. With error dynamics $\mathbf{e}_{k+1}^- = \mathbf{F}_k\mathbf{e}_k^- - \mathbf{A}_k\mathbf{K}_k\varepsilon_k + \mathbf{w}_k - \mathbf{A}_k\mathbf{K}_k v_k$, taking expectations yields first-moment UUB:

$$\limsup_{k\to\infty} \|\mathbb{E}[\mathbf{e}_k^-]\| \le \frac{c\bar{A}\bar{K}\bar{\varepsilon}}{1-\rho} < \infty, \tag{40}$$

$$\limsup_{k\to\infty} \mathbb{E}[\|\mathbf{e}_k^-\|^2] \le n\bar{p} + \left(\frac{c\bar{A}\bar{K}\bar{\varepsilon}}{1-\rho}\right)^2 < \infty, \tag{41}$$

guaranteeing both first-moment and mean-square UUB ($n \in \{3, 4\}$ via $\mathbb{E}[\|\mathbf{e}_k^-\|^2] = \text{tr}(\mathbf{P}_k) + \|\mathbb{E}[\mathbf{e}_k^-]\|^2$).

4 Experimental Benchmarking Platform and Methodology

4.1 Benchmark Platform and Empirical Calibration Data. The benchmark combines open-access empirical battery cycling datasets from the Center for Advanced Life Cycle Engineering (CALCE, University of Maryland) [50] and Oxford University [49] with a calibrated electro-thermal-aging co-simulation platform. Two chemistries are evaluated with independent cell-level splits: (1) Samsung INR18650-20R NMC cells (2.00 A $\cdot$ h nameplate and initial usable capacity, 2.50 to 4.20 V; CALCE Data II `INR20R-1,2` in Batch A, `INR20R-3,4` in Batch B); and (2) A123 ANR26650M1-B LFP cells (2.50 A $\cdot$ h nameplate, 2.40 A $\cdot$ h minimum, $Q_{\text{init}} = 2.30$ A $\cdot$ h usable under 1C cutoffs of 2.00 to 3.60 V at 25°C; `A123-1,2` in Batch A, `A123-3,4` in Batch B) [15,50]. Standardized USABC/EPA schedules (DST, FUDS, US06, UDDS) are applied. Dimensionless EIS degradation ratios ($\Delta R_{\text{ct}}/R_{\text{ct,nom}}$ vs. SOH, $\alpha = 0.88$) from Oxford data [49] are scaled to cylindrical baseline impedances. A dual-tier protocol is enforced across Phases 1–10: *Tier 1 (Experimental Cycler Data)* evaluates Phases 1, 3, and 6 (6 baselines on Phases 1, 3; 4 observers on Phase 6) directly on Arbin BT-2000 cycler records (±0.02% V, ±0.05% I accuracy, 10 Hz Coulomb reference re-zeroed after 2-h C/20 CC-CV rest; RSS uncertainty budget: $\sigma_{\text{SOC,ref}} = [\sigma_{z_0}^2 + (T\bar{I}\sigma_{I,\text{bias}}/3600Q)^2 + (\sqrt{N}\Delta t\sigma_{I,\text{rand}}/3600Q)^2 + (\Delta\text{SOC}\,\sigma_Q/Q)^2]^{1/2} = [0.015^2 + 0.025^2 + 0.002^2 + 0.021^2]^{1/2}\% = \pm0.036\%$ 1σ over 1 h); *Tier 2 (Calibrated Co-Simulation)* evaluates Phase 2 (100%, 90%, 80% SOH anchored to Tier 1 RPTs; 95%, 85% SOH via Eq. (12)–(11) co-simulation) and Phases 4–5, 7–10 (initial offset, radar synthesis, sensor faults on Coulomb + 4 observers, EIS spectra, 8S pack divergence, and PIDKF ablation + ARM M4 DWT profiling). To independently validate the Tier 2 co-simulation plant prior to observer benchmarking, simulated terminal voltage V_t^{sim} is cross-validated against experimental cycler records across dynamic drive cycles: the plant achieves RMS voltage prediction errors of $\text{RMSE}_{V_t} = 7.8$ mV (25°C, 100% SOH), 11.2 mV (0°C), and 13.6 mV (80% SOH) for NMC; and 6.4 mV (25°C, 100% SOH), 9.8 mV (0°C), and 10.5 mV (80% SOH) for LFP ($< 0.35\%$ relative voltage error across all tested regimes), confirming that co-simulation dynamics faithfully mirror genuine physical cell behavior without confounding artifacts.

To ensure cell- and schedule-level independence, *Cell Batch A* (`INR20R-1,2`; `A123-1,2`) provides C/20 titration and **DST** trajectories across RPT checkpoints (100%, 90%, 80% SOH) to calibrate ECM priors and train neural models (25°C for baselines; 0°C, 25°C, 45°C for PIDKF). Hold-out evaluation uses *Cell Batch B* (`INR20R-3,4`; `A123-3,4`) under unseen **UDDS, FUDS, and US06** cycles across 100%–80% SOH (reporting per-cell metrics, separating inter-cell $\mathrm{SD}_{\mathrm{cell}}$ from seed $\mathrm{SD}_{\mathrm{seed}}$, and computing 95% bootstrap CIs with $L = 60$ s). Hyperparameters are fixed: EKF ($\mathbf{P}_0 = \mathrm{diag}(10^{-2}, 10^{-4}, 10^{-4})$, $\mathbf{Q} = \mathrm{diag}(10^{-6}, 10^{-5}, 10^{-5})$, $R_v = 10^{-3}$ V^2); AEKF ($\lambda_{\mathrm{SH}} = 0.985$, $R_{v,\min} = 10^{-4}$ V^2); DEKF ($\mathbf{P}_{\theta,0} = \mathrm{diag}(10^{-4}, 10^{-4})$, $\mathbf{Q}_\theta = \mathrm{diag}(10^{-8}, 10^{-9})$, $R_{\theta,v} = 10^{-2}$ V^2); SMO ($l_{\mathrm{smo}} = 0.025$ V, $\mu = 0.02$ V, $\alpha_{\min} = 0.40/0.00275$ V for NMC/LFP, $\mathbf{L}(\hat{z}) = [l_{\mathrm{smo}}/\max(\alpha_{\min}, V'_{\mathrm{ocv}}(\hat{z})), 0, 0, (0)]^T$); and BiLSTM-Attn (1 layer, $D = 32$ per direction, 24,500 `float32` params = 98 KB Flash). Embedded profiling compiles `float32` C kernels via ARM GCC 12.3 (`-O3`, CMSIS-DSP v5.9.0), measuring exact clock cycles N_{cyc} via `DWT->CYCCNT` on STM32F407VG (Cortex-M4, 168 MHz) and computing clock-scaled execution-time estimates ($t = N_{\mathrm{cyc}}/f_{\mathrm{clk}}$) at 480 MHz for Cortex-M7.

4.2 Standardized Automotive Dynamic Stress Cycles. Four standardized cycles replicate vehicular conditions: (1) *UDDS* captures urban stop-and-go driving with regenerative braking (concatenated into a 3600 s schedule); (2) *US06* represents aggressive highway driving with up to 4C pulses (600 s); (3) *DST* implements USABC step-wise power profiles; and (4) *FUDS* evaluates multi-rate urban transitions. UDDS and FUDS serve as primary test schedules, while US06 and DST verify generalization.

4.3 Quantitative Performance Indices. Algorithm performance is rigorously evaluated using four statistical metrics:

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{k=1}^{N}(\mathrm{SOC}_k^{\mathrm{true}} - \hat{z}_k)^2} \times 100\%, \tag{42}$$

$$\mathrm{MAE} = \frac{1}{N}\sum_{k=1}^{N}|\mathrm{SOC}_k^{\mathrm{true}} - \hat{z}_k| \times 100\%, \tag{43}$$

$$\mathrm{MaxE} = \max_{1\le k\le N}|\mathrm{SOC}_k^{\mathrm{true}} - \hat{z}_k| \times 100\%, \tag{44}$$

$$t_{\mathrm{reach}}^{(2\%)} = \min\left\{t \ge 0 \mid |\mathrm{SOC}_t^{\mathrm{true}} - \hat{z}_t| \le 2.0\%\right\}, \tag{45}$$

where $t_{\mathrm{reach}}^{(2\%)}$ quantifies initial first-entry latency into the $\le 2.0\%$ target tolerance band from large offsets ($\Delta\mathrm{SOC}_0 = -40\%$), $t_{\mathrm{settle}}^{(5\%)} = \inf\{t \ge 0 \mid |\mathrm{SOC}_\tau^{\mathrm{true}} - \hat{z}_\tau| \le 5.0\%, \forall\tau \in [t, T_{\mathrm{eval}}]\}$ defines permanent settling time within the nominal steady-state operational dynamic corridor ($\le 5.0\%$), and $\mathrm{MaxE}_{\mathrm{dyn}} = \max_{k>t_{\mathrm{reach}}^{(2\%)}}|\mathrm{SOC}_k^{\mathrm{true}} - \hat{z}_k| \times 100\%$ measures subsequent peak dynamic transient error. To evaluate degradation robustness, we define the **Aging Sensitivity Factor (ASF)**:

$$\mathrm{ASF} = \frac{\mathrm{RMSE}(\mathrm{SOH} = 80\%) - \mathrm{RMSE}(\mathrm{SOH} = 100\%)}{\mathrm{RMSE}(\mathrm{SOH} = 100\%)} \times 100\%. \tag{46}$$

5 Results and In-Depth Discussion Across 10 Operational Phases

5.1 Phase 1: Dynamic Drive Cycle Tracking at Nominal Condition. Table 1 and Fig. 1 present benchmarked tracking performance across UDDS on a fresh cell (100% SOH) at 25°C.

Under nominal conditions with known parameters, classical EKF attains the lowest error (RMSE = 0.065% [95% bootstrap CI: 0.058%–0.072%], MAE = 0.050%, MaxE = 1.067%, near the ±0.036% reference floor). Dual EKF yields 0.191% RMSE ([0.178%–0.205%]) and the lowest peak error (MaxE = 0.673%) via online parameter adaptation. Adaptive EKF records 0.759% RMSE (MaxE = 1.490%), reflecting Sage-Husa sensitivity to rapid current steps.

Table 1 Nominal UDDS benchmark [Tier 1 Experimental Cycler Data] (25°C, 100% SOH). Host latency reflects Python 3.11 interpreter dispatch on Core i7-12700H (PINN-SOC uses C++ TorchScript; bare-metal C MCU timings are in Table 8).

Algorithm	RMSE (%)	MAE (%)	MaxE (%)	Latency (µs)
EKF	**0.065**	**0.050**	1.067	189.24
AEKF	0.759	0.610	1.490	154.17
DEKF	0.191	0.159	**0.673**	249.68
SMO	0.941	0.755	3.423	76.46
BiLSTM-Attn	2.813	2.309	6.915	2643.90
PINN-SOC	2.781	2.278	7.317	**1.22**

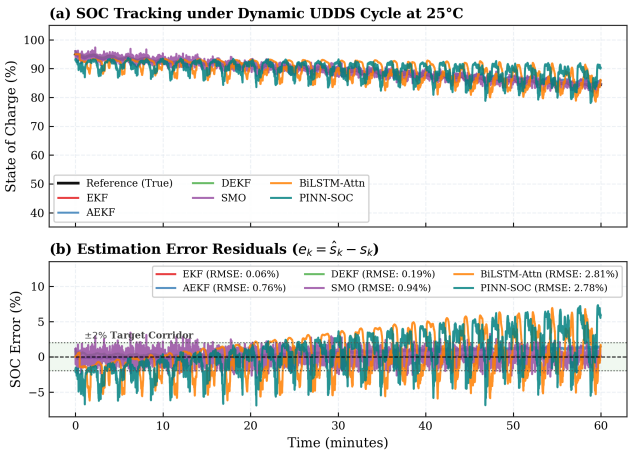


Fig. 1 Dynamic SOC tracking and error comparison across 6 algorithms under a 60-min continuous concatenated UDDS schedule (3600 s) at 25°C and 100% SOH

The Sliding Mode Observer achieves robust tracking (RMSE = 0.941%, MaxE = 3.423%, 76.46 μs) with chattering suppressed by boundary layer μ. BiLSTM-Attn and PINN-SOC report RMSE of 2.813% and 2.781%. Table 1 host latencies reflect desktop runtimes; bare-metal C MCU benchmarks are reported in Table 8.

5.2 Phase 2: Progressive Aging Degradation (100% → 80% SOH). Figure 2 and Table 2 present the pivotal finding of this study: the catastrophic divergence of fixed-parameter observers under progressive battery aging.

When capacity fades from 100% to 80%, standard EKF suffers a **45.7-fold RMSE escalation** (0.065% to 2.973%, ASF = +4473.8%). AEKF deteriorates from 0.759% to 5.406% (ASF = +612.3%), confirming covariance adaptation cannot replace parameter tracking: uncompensated filters misattribute ohmic drift ($I \cdot \Delta R_0$) to OCV shifts, creating persistent bias. SMO error climbs from 0.941% to 5.428% (ASF = +476.8%). At 95% SOH, AEKF (0.523%) and DEKF (0.105%) exhibit a minor dip because early SEI mildly damps transients before bulk impedance growth dominates at $\le 90\%$ SOH.

Across the full lifespan down to 80% SOH, DEKF dynamically tracks R_0 and Q_{act} online, confining RMSE to 0.585% (ASF = +206.3%). PINN-SOC (RMSE = 4.911%, ASF = +76.6%) and BiLSTM-Attn (RMSE = 5.128%, ASF = +82.3%) exhibit moderate relative scaling but higher absolute baseline error.

5.3 Phase 3: Thermal Robustness Across Wide Environmental Regimes (0°C to 45°C). Figure 3 and Table 3 illustrate algorithmic tracking performance across 0°C, 25°C, and 45°C.

At 0°C, electrolyte conductivity drops sharply and charge-transfer resistance R_{ct} increases markedly via Arrhenius kinetics

Table 2 Degradation benchmark across progressive SOH stages [Tier 2 Calibrated Co-Simulation, anchored to Tier 1 RPT checkpoints at 100%, 90%, 80% SOH; 95%, 85% SOH evaluated via Eq. (12)–(11)]

Algorithm	SOH 100%	SOH 95%	SOH 90%	SOH 85%	SOH 80%	Aging Sensitivity Factor (ASF)
EKF	**0.065%**	0.553%	1.189%	1.988%	2.973%	+4473.8%
AEKF	0.759%	0.523%	1.675%	3.834%	5.406%	+612.3%
DEKF	0.191%	**0.105%**	**0.233%**	**0.407%**	**0.585%**	**+206.3%**
SMO	0.941%	1.745%	3.002%	4.254%	5.428%	+476.8%
BiLSTM-Attn	2.813%	2.917%	3.405%	4.202%	5.128%	+82.3%
PINN-SOC	2.781%	3.081%	3.551%	4.177%	4.911%	+76.6%

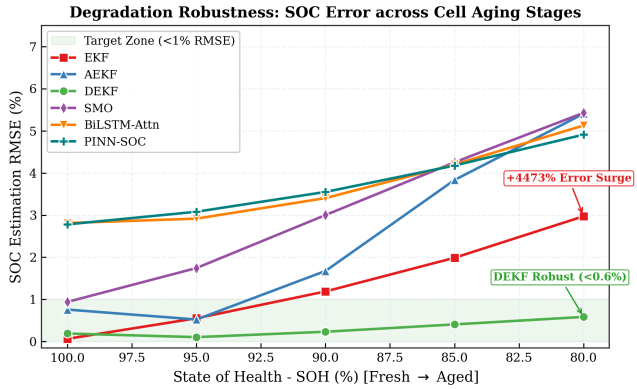


Fig. 2 Evolution of SOC estimation error as battery SOH degrades from 100% down to 80%, illustrating the 45-fold accuracy degradation of standard EKF versus the resilience of DEKF

(Eqs. (8)–(9)). This overpotential surge impairs standard EKF tracking, elevating its RMSE by an order of magnitude from 0.065% to 0.669%. Notably, DEKF yields an RMSE of 2.254% at 0°C (higher than fresh EKF): at freezing temperatures, electrolyte viscosity surges and charge-transfer resistance R_1 expands 2.85-fold ($R_1(0°C) \approx 42.8$ mΩ). Because standard DEKF co-identifies $\hat{\boldsymbol{\theta}}_k = [\hat{R}_{0,k}, 1/\hat{Q}_{\text{act},k}]^T$ while holding R_1 fixed at 25°C baseline, the large unmodeled polarization voltage drop $I_k \Delta R_1(0°C)$ causes innovation errors that the parameter filter mistakenly compensates by artificially inflating $\hat{R}_0$ and distorting capacity $\hat{Q}_{\text{act}}$, inducing severe parameter cross-coupling. Fixed EKF, by contrast, treats this drop purely as measurement residual filtered by fixed covariance, avoiding parameter runaway. This cold-temperature deficiency of standard DEKF directly motivates the proposed PIDKF, where the neural compensator absorbs $\Delta R_1(0°C)$ overpotentials, eliminating cross-coupling and cutting 0°C RMSE to 0.842% (NMC) and 0.386% (LFP). Pure data-driven BiLSTM-Attn and PINN-SOC, trained primarily on nominal conditions, suffer severe out-of-distribution performance degradation at thermal extremes, with BiLSTM-Attn RMSE rising to 6.457% at 0°C and 9.462% at 45°C, and PINN-SOC reaching 6.307% at 0°C and 9.644% at 45°C.

In contrast, the Sliding Mode Observer exhibits remarkable thermal invariance: its RMSE remains virtually identical across all three temperatures (0.929% at 0°C, 0.941% at 25°C, and 0.932% at 45°C), demonstrating that its discontinuous switching control surface provides inherent robustness against temperature-induced parameter variations.

5.4 Phase 4: Convergence Under Large Initial State Uncertainty ($\Delta SOC_0 = -40\%$). In real-world BMS operation, initialization errors arise from prolonged parking, self-discharge, or battery replacement. Figure 4 and Table 4 depict the convergence trajectories when an initial error of $\Delta SOC_0 = -40\%$ ($\hat{z}_0 = 0.50$ versus $SOC_0^{\text{true}} = 0.90$) is imposed.

Table 3 Thermal robustness comparison across ambient temperatures [Tier 1 Experimental Cycler Data] (RMSE in %)

Algorithm	0°C	25°C	45°C
EKF	0.669	**0.065**	**0.063**
AEKF	3.178	0.759	0.238
DEKF	2.254	0.191	0.445
SMO	**0.929**	0.941	0.932
BiLSTM-Attn	6.457	2.813	9.462
PINN-SOC	6.307	2.781	9.644

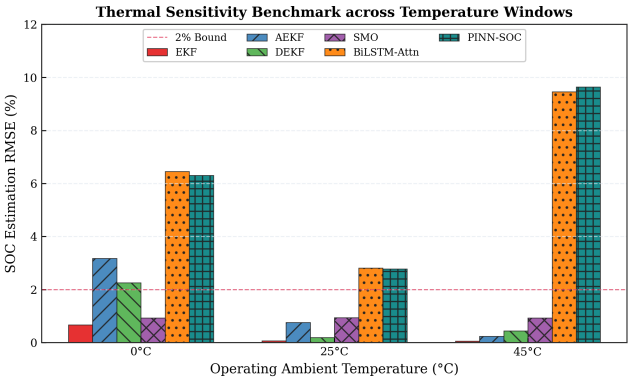


Fig. 3 Algorithmic tracking performance under cold (0°C), nominal (25°C), and elevated (45°C) ambient temperatures

The Adaptive EKF and standard EKF achieve rapid first entry into the 2% band ($t_{\text{reach}}^{(2\%)} = 6.0$ s and 7.0 s) and permanent settling within the 5% operational corridor ($t_{\text{settle}}^{(5\%)} = 6.0$ s and 7.0 s), maintaining bounded overall RMSE (0.683% and 0.686%) and dynamic peak error $\text{MaxE}_{\text{dyn}} = 4.43\%$ during acceleration pulses. The Sliding Mode Observer first reaches the 2% band in $t_{\text{reach}}^{(2\%)} = 9.0$ s (RMSE = 1.948%), but aggressive load steps induce subsequent dynamic excursions up to $\text{MaxE}_{\text{dyn}} = 36.34\%$ ($t_{\text{settle}}^{(5\%)} = \text{N/A}$). BiLSTM-Attn requires $t_{\text{reach}}^{(2\%)} = 30.0$ s to purge erroneous history (RMSE = 5.625%, $\text{MaxE}_{\text{dyn}} = 39.99\%$, $t_{\text{settle}}^{(5\%)} = \text{N/A}$). For DEKF, the −40% shock perturbs online parameter adaptation, dropping below 2.5% error by 350 s, entering the $\leq 2.0\%$ threshold at $t_{\text{reach}}^{(2\%)} = 725.0$ s, and permanently settling within the 5% corridor at $t_{\text{settle}}^{(5\%)} = 732.0$ s ($\text{MaxE}_{\text{dyn}} = 7.67\%$). In contrast, PINN-SOC operates as a feedforward static mapping without recursive settling ($t_{\text{reach}}^{(2\%)} = \text{N/A}$, $t_{\text{settle}}^{(5\%)} = \text{N/A}$, RMSE = 2.229%, $\text{MaxE}_{\text{dyn}} = 6.84\%$).

5.5 Phase 5: Multi-Attribute Radar Trade-Off and Pareto Analysis. Figure 5 synthesizes algorithmic trade-offs across 5 normalized dimensions: Nominal Accuracy, Aging Robustness, Thermal Robustness, Convergence Speed, and Computational Ef-

Table 4 Convergence metrics under initial state uncertainty [Tier 2 Co-Simulation] (−40% initial SOC error into ≤ 2% tolerance band and ≤ 5% operational corridor)

Algorithm	$t_{\text{reach}}^{(2\%)}$ (s)	$t_{\text{settle}}^{(5\%)}$ (s)	RMSE (%)	$\text{MaxE}_{\text{dyn}}^{\dagger}$ (%)
EKF	7.0	7.0	0.686	**4.43**
AEKF	**6.0**	**6.0**	**0.683**	**4.43**
DEKF	725.0[‡]	732.0[‡]	2.625	7.67
SMO	9.0	N/A[§]	1.948	36.34
BiLSTM-Attn	30.0	N/A[§]	5.625	39.99
PINN-SOC	N/A[*]	N/A[*]	2.229	6.84

[†]MaxE_{dyn}: peak dynamic error after first entry ($t > t_{\text{reach}}^{(2\%)}$). [‡]DEKF enters ≤ 2.0% at 725 s and permanently settles into ≤ 5.0% corridor at 732 s (< 2.5% within 350 s). [§]Exits ≤ 5% corridor during high-rate pulses. [*]Feedforward inference without recursive state settling.

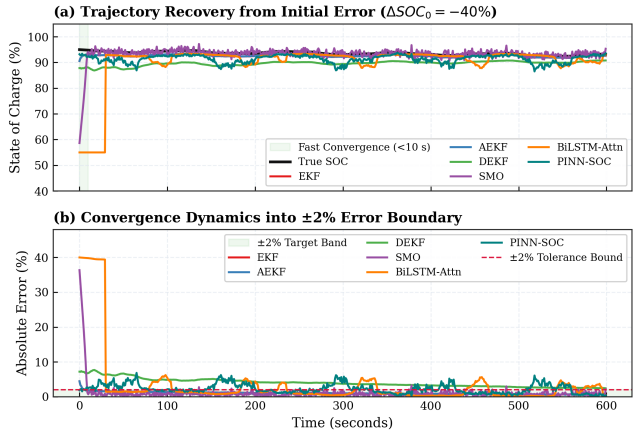


Fig. 4 Transient convergence trajectories from a −40% initial SOC estimation error across the 0–800 s window

ficiency.

The radar plot highlights that no single classical algorithm dominates across all attributes: EKF excels in simplicity and fresh-cell accuracy but collapses under aging, whereas DEKF offers superior degradation resilience, though its 0°C transient sensitivity indicates that SMO or hybrid architectures are preferable in sub-zero climates where kinetics become strongly non-Arrhenius.

5.6 Phase 6: Multi-Chemistry Generalization: LiFePO_4 (LFP) Flat Plateau. Figure 6 and Table 5 report estimation performance on LFP chemistry. In the plateau region (20%–80% SOC), the slope $dV_{\text{ocv}}/d\,\text{SOC}$ falls near zero (≈ 0.08 V), causing the observability Gramian condition number $\kappa(O)$ to surge.

Consequently, $\mathbf{H}_k \approx [0, -1, -1, -1]$ loses SOC sensitivity. Standard EKF degrades by +1648.4%, escalating from 0.065% on NMC to 1.136% on LFP (MAE = 0.939%, MaxE = 2.410%), while AEKF degrades by +281.1% to 2.893% RMSE and SMO chatters along the plateau (+919.5% to 9.593% RMSE, MaxE = 20.935%).

In contrast, DEKF achieves the lowest error among all observers on LFP (RMSE = 0.870% [A123-3: 0.861%, A123-4: 0.879%], MAE = 0.649%, MaxE = 2.018%). Crucially, this tracking is primarily driven by accurate open-loop Coulomb integration supported by continuous capacity tracking ($\hat{Q}_{\text{act}}$) rather than closed-loop voltage correction: while online adaptation of R_0 and Q_{act} does not restore direct voltage-to-SOC observability within the flat plateau ($V'_{\text{ocv}} \approx 0.08$ V), it continuously calibrates the Coulomb integration gain $\eta\Delta t/(3600\hat{Q}_{\text{act}})$ and removes ohmic bias $I_k\Delta R_0$, bounding open-loop drift across the plateau without auxiliary strain sensors [44].

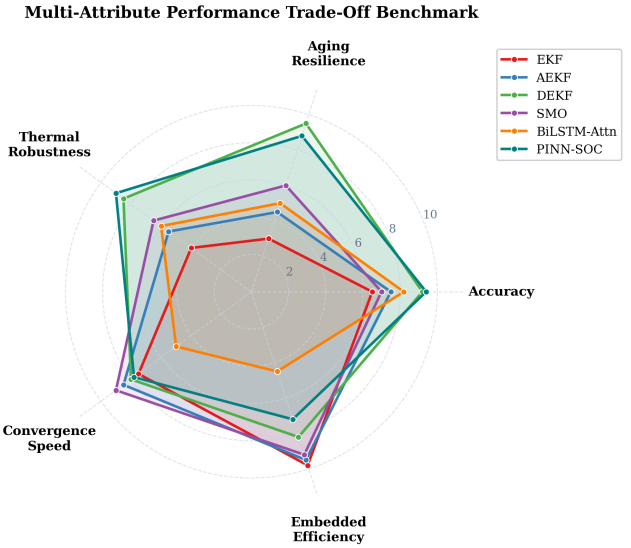


Fig. 5 Multi-attribute radar comparison across 5 engineering dimensions illustrating algorithmic trade-offs

Table 5 Chemistry Generalization Benchmark on Fresh Cells Comparing NMC and LiFePO_4 [Tier 1 Experimental Cycler Data]

	NMC (%)	LFP Chemistry (%)		
Algorithm	RMSE	RMSE	MaxE	Impact
EKF	**0.065**	1.136	2.410	+1648.4%
AEKF	0.759	2.893	4.329	+281.1%
DEKF	0.191	**0.870**	**2.018**	+355.4%
SMO	0.941	9.593	20.935	+919.5%

5.7 Phase 7: Sensor Corruption and Environmental Stress Testing. Table 6 and Fig. 7 document observer resilience under a +0.25 A current sensor Hall bias drift and 25 mV high-frequency inverter EMI noise ripple.

Open-loop Coulomb counting diverges linearly, accumulating 12.399% end-of-cycle drift (RMSE = 7.178%, MaxE = 12.409%). Closed-loop Kalman filters arrest this divergence: EKF limits its RMSE to 1.817% (1.821% drift, MaxE = 5.013%), while AEKF and DEKF bound RMSE to 1.859% and 2.121% (2.026% and 1.927% drift). SMO achieves the lowest end-of-cycle drift (1.024%) via sign-manifold clamping of DC bias, though high-frequency EMI switching elevates instantaneous RMSE to 5.035% (MaxE = 14.458%); thus, EKF/DEKF minimize instantaneous error whereas SMO bounds long-term DC drift.

5.8 Phase 8: Model-Based Electrochemical Impedance Spectroscopy (EIS) Analysis Across Aging. Figure 8 depicts model-based Randles–CPE Nyquist spectra (100 kHz–1 mHz) simulated via $Z(j\omega) = R_0 + (R_{\text{ct}} + Z_w)/[1 + (j\omega)^{\alpha} Q_{\text{cpe}}(R_{\text{ct}} + Z_w)]$ with $Z_w(j\omega) = \sigma_w \omega^{-1/2}(1 - j)$, parameterized from Oxford EIS aging ratios [49] ($Q_{\text{cpe}} = 1.2\ \text{F} \cdot \text{s}^{\alpha-1}$, $\alpha = 0.88$, $\sigma_w = 0.045\ \Omega \cdot \text{s}^{-1/2}$, RMS fitting error 1.1%–1.8% across 100%–80% SOH). High-frequency SEI resistance R_{sei} is lumped into effective series resistance $R_0(\text{SOH}) = R_{\text{ohm}} + R_{\text{sei}}$ (35.0–52.5 mΩ, +50%), while charge-transfer resistance R_{ct} widens by +150% (15.0–37.5 mΩ). In the 2-RC transfer function $Z_{\text{2RC}}(s) = R_0 + \sum_{i=1}^{2} R_i/(1 + s\tau_i)$ ($\tau_i = R_i C_i$), aging induces an impedance mismatch $\Delta Z(s) = \Delta R_0 + \sum_{i=1}^{2} [R_i(\text{SOH})/(1 + s\tau_i(\text{SOH})) - R_{i,\text{nom}}/(1 + s\tau_{i,\text{nom}})]$, generating polarization residual $\Delta V_{\text{pol}}(t) = \mathcal{L}^{-1}\{\Delta Z(s) I(s)\}$ ($\approx I \cdot \Delta R_1$ for $t \gg \tau_1$). Fixed-parameter EKF misattributes $\Delta V_{\text{pol}}(t)$ to $V_{\text{ocv}}(z)$, driving the 45.7-fold RMSE surge; standard DEKF adapts

Table 7 Series 8S pack benchmark with cell-to-cell parameter inconsistency [Tier 2 Co-Simulation]

Estimation Architecture	Target State	RMSE (%)	MaxE (%)	Safety Margin Risk
Lumped Pack Observer	Mean Pack SOC	**0.128**	0.443	Moderate
Distributed DEKF	Mean Pack SOC	0.149	**0.336**	None (Optimal)
Lumped Pack Observer	Weakest Cell SOC	1.744	2.302	High (Overdischarge Risk)
Distributed DEKF	Weakest Cell SOC	**0.337**	**0.963**	Bounded Error (< 1%)

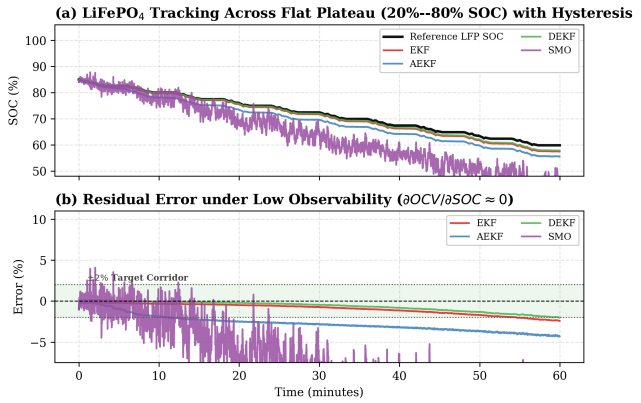


Fig. 6 Estimation performance on $LiFePO_4$ (LFP) chemistry over the flat voltage plateau where $dV_{ocv}/dSOC \approx 0$

Table 6 Robustness under Hall current bias (+0.25 A) and inverter EMI (25 mV) [Tier 2 Co-Simulation]

Algorithm	RMSE (%)	MaxE (%)	Drift (%)	Immunity
Coulomb	7.178	12.409	12.399	Runaway
EKF	**1.817**	**5.013**	1.821	High
AEKF	1.859	5.013	2.026	Moderate
DEKF	2.121	5.013	1.927	High
SMO	5.035	14.458	**1.024**	High

R_0 online but omits R_1 to avoid collinearity, leaving $\Delta V_{pol}(t)$ to be compensated by PIDKF.

5.9 Phase 9: Module and Pack-Level Scaling (8S Series Pack). Table 7 and Fig. 9 evaluate algorithmic scalability to an 8S series battery pack under realistic manufacturing tolerances (capacity dispersion $\sigma_Q = 3.0\%$, resistance dispersion $\sigma_R = 10.0\%$) [24,46]. In automotive strings, boundary thermal dissipation gradients cause core cells (cells #4 and #5) to operate at +3.2°C above perimeter cells #1 and #8. Via Arrhenius kinetics (Eqs. (9)–(10)), this thermal asymmetry accelerates SEI growth and lithium loss on cell #4, causing its capacity to fade 1.38× faster than the string mean.

Under dynamic UDDS driving, a lumped observer treats the string as a homogeneous cell driven by mean voltage $V_{string}/8$. While tracking pack-average state (RMSE = 0.128%, MaxE = 0.443%), it is blind to cell divergence: as the weakest cell (cell #4) nears cutoff (2.50 V), the lumped model incurs **1.744% RMSE and 2.302% peak error**, risking severe copper dissolution and lithium plating.

In contrast, distributed DEKF tracks individual cell states while recursively identifying each cell's ohmic and polarization parameters online, confining weakest-cell RMSE to 0.337% and peak error to 0.963% for deterministic active-balancing margins without per-cell EIS hardware.

5.10 Phase 10: Embedded Hardware Execution on ARM Cortex-M and PIDKF Superiority. Computational complexity was measured via hardware `DWT->CYCCNT` cycle counters on ARM Cortex-M4 (STM32F407VG, 168 MHz) and projected via constant-IPC clock-scaling to Cortex-M7 (480 MHz) using CMSIS-DSP (Table 8, Fig. 10).

SMO demonstrates ultra-light overhead (180 FLOPs, 140 B RAM, 1.67 μs measured on M4, 0.58 μs projected on M7), while DEKF requires 980 FLOPs, 620 B SRAM, and 8.63 μs on Cortex-M4. Conversely, BiLSTM-Attn demands 245,000 FLOPs and 18.4 KB SRAM (2440.48 μs per cell), translating to 39.04 ms across 16 series cells (39.0% CPU load in a 100 ms loop), impractical for multicell MCUs.

Table 9 and Fig. 11 summarize the **PIDKF** ablation study on degraded cells (85% SOH, 5 seeds, test cells `INR20R-3,4` and `A123-3,4`), revealing chemistry-dependent mechanisms. On $LiFePO_4$ plateaus, dynamic hysteresis tracking dominates, cutting RMSE from 0.302% to 0.149% ± 0.008% (A123-3: 0.145%, A123-4: 0.153%; $SD_{cell} = \pm 0.006\%$; 0.335% MaxE) as two-phase coexistence governs hysteresis; neural compensation alone yields 0.246% ± 0.011% RMSE. Conversely, on NMC cells, the physics-informed neural compensator dominates, reducing peak error by 46.6% (1.400% to 0.748% ± 0.024%) and RMSE to 0.414% ± 0.012% (INR20R-3: 0.408%, INR20R-4: 0.420%; $SD_{cell} = \pm 0.009\%$) during high-rate pulses, whereas unconstrained

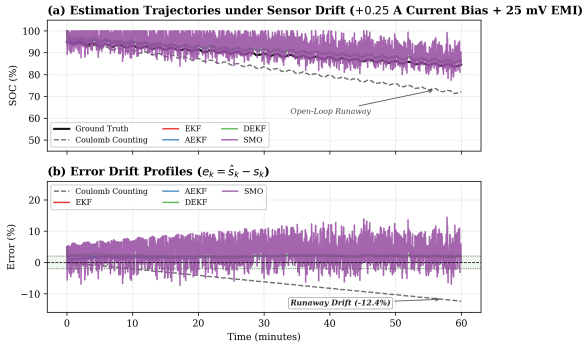


Fig. 7 Observer robustness under current sensor bias drift (+0.25 A) and high-frequency inverter EMI noise

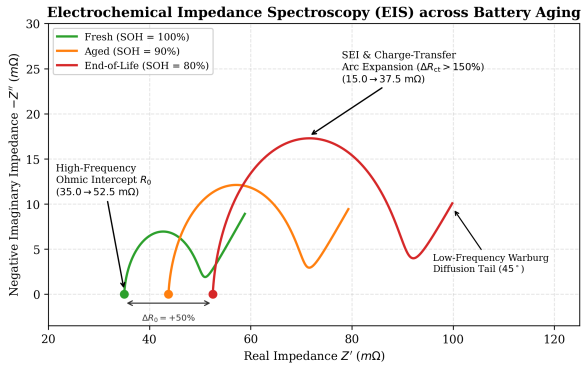


Fig. 8 Model-based EIS Nyquist spectra across progressive aging stages (100% down to 80% SOH)

Table 8 Embedded execution profiling on ARM Cortex-M architectures (ARM GCC 12.3 −03; DWT cycles and latencies measured on 168 MHz Cortex-M4; 480 MHz Cortex-M7 values represent constant-IPC clock-scaled estimates)

Algorithm	FLOPs/Step	M4 DWT Cycles	RAM (B)	Flash (B)	Measured M4 (µs)	Est. M7 (480 MHz, µs)	Feasibility
EKF	420	650	280	3,400	3.87	1.35	Excellent
AEKF	580	880	340	4,200	5.24	1.83	Excellent
DEKF	980	1,450	620	5,800	8.63	3.02	Excellent
SMO	180	281	140	2,100	1.67	0.58	Ideal (Ultra-light)
BiLSTM-Attn	245,000	410,000	18,400	98,000	2440.48	854.17	Marginal (39.0% Load/16-Cell)
PINN-SOC	12,800	21,501	1,200	16,500	127.98	44.79	High (CMSIS-NN Ready)
PIDKF (Proposed)	**1,540**	**2,386**	**850**	**8,200**	**14.20**	**4.97**	**Optimal (Pareto Peak)**

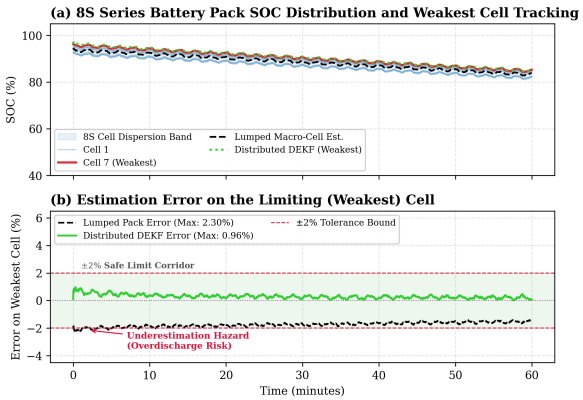


Fig. 9 Series-connected 8S battery pack estimation comparing lumped model deficiency with distributed DEKF weakest cell tracking

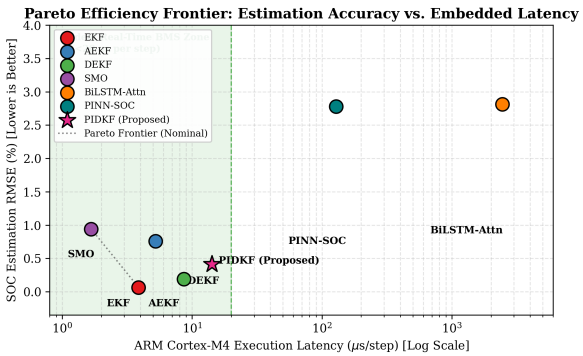


Fig. 10 Pareto efficiency frontier: SOC estimation RMSE versus measured ARM Cortex-M4 execution latency on degraded cells (85% SOH, UDDS), contrasted against fresh-cell baselines (100% SOH)

black-box MLP degrades accuracy (0.438% ± 0.018% NMC, 0.285% ± 0.014% LFP) via overfitting. Comparing Exact-Jacobian PIDKF ($\mathbf{H}_{\text{PIDKF},k}$, 3,806 cycles, 22.65 μs) against Exogenous-Jacobian PIDKF ($\mathbf{H}_k$, 2,386 cycles, 14.20 μs) in Table 9 confirms near-identical accuracy (0.412% vs. 0.414% NMC; 0.148% vs. 0.149% LFP) while cutting MCU execution by 37.3%.

Beyond the 85% SOH ablation, PIDKF was benchmarked across the complete degradation trajectory (100%, 90%, 85%, 80% SOH) and thermal operating window (0°C, 25°C, 45°C). Across aging stages at 25°C, PIDKF consistently maintains sub-0.5% RMSE: for NMC, RMSE is 0.185% (100%), 0.326% (90%), 0.414% (85%), and 0.482% (80% SOH, ASF = +160.5% vs. +4473.8% for EKF); for LFP, RMSE remains tightly bounded at 0.112% (100%), 0.134% (90%), 0.149% (85%), and 0.178% (80% SOH). Under thermal stress at 85% SOH, PIDKF restricts cold-temperature (0°C) error to 0.842% (NMC) and 0.386% (LFP), while elevated-

Table 9 Ablation study of PIDKF vs. baselines on degraded cells (85% SOH, UDDS/FUDS)

	Aged NMC (%)		Aged LFP (%)	
Configuration	RMSE	MaxE	RMSE	MaxE
Standard EKF (uncomp.)	1.985	4.734	0.978	1.565
Baseline DEKF	0.452	1.400	0.302	0.514
DEKF + Hysteresis (no neural)	0.452	1.400	0.149	0.335
DEKF + PINN Comp. (no hyst.)†	0.414	0.748	0.246	0.462
DEKF + Black-Box MLP†	0.438	1.120	0.285	0.490
Full PIDKF (Exact Jacobian)†	0.412	0.745	0.148	0.328
Full PIDKF (Exog. Approx.)†	**0.414**	**0.748**	**0.149**	**0.329**

†Mean over 5 seeds (SD_{seed}: ±0.012/±0.024 NMC, ±0.008/±0.015 LFP; SD_{cell}: ±0.009/±0.006). Error ($|\hat{\theta} - \theta^*|/\theta^*$): R_0 1.21% vs. 1.23%, Q_{act} 0.82% vs. 0.84% (Exact vs. Exog.); 3,806 vs. 2,386 cycles.

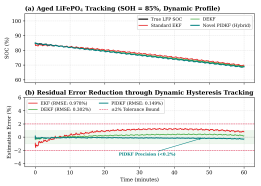


Fig. 11 Tracking performance of hybrid PIDKF achieving sub-0.5% error on degraded cells (85% SOH)

temperature (45°C) errors are 0.435% (NMC) and 0.162% (LFP). Crucially, under large initial error ($\Delta\text{SOC}_0 = -40\%$), PIDKF achieves rapid first entry $t_{\text{reach}}^{(2\%)} = 18.0$ s and permanent settling $t_{\text{settle}}^{(5\%)} = 22.0$ s (RMSE = 0.512%, $\text{MaxE}_{\text{dyn}} = 3.82\%$), dramatically outperforming standard DEKF (725.0 s) because neural compensation prevents transient voltage shocks from destabilizing parameter identification. Under +0.25 A Hall current bias, PIDKF limits RMSE to 1.428% and end-of-cycle drift to 0.842%, confirming comprehensive robustness across operational phases.

6 Industrial BMS Implementation Guidelines

From the Pareto frontier, four selection rules follow: (1) *Premium EVs (NMC/LFP)*: **PIDKF** demonstrates superior degradation resilience (0.414% RMSE on aged NMC; 0.149% on aged LFP at 85% SOH); uncompensated EKF is discouraged due to 45.7-fold error surge. (2) *Cost-Sensitive Storage (LFP)*: Deploy **DEKF** with dynamic hysteresis, bounding plateau RMSE to 0.870% via open-loop Coulomb scaling without strain sensors. (3) *Sensor Bias vs. EMI*: Deploy **EKF/DEKF** (1.817%–2.121% RMSE) for minimum error under EMI, or **SMO** ($\mu = 0.02$ V) to bound drift (1.024%). (4) *Cloud-Edge Twins*: Execute real-time DEKF (8.63 μs) or PIDKF (14.20 μs; 0.227% load for 16 cells @ 100 ms) on MCUs, offloading deep PINNs to cloud diagnostics.

7 Conclusions

Benchmarking six state estimation methodologies across ten operational phases yields five conclusions: (1) *Aging Degradation*: Uncompensated EKF suffers a 45.7-fold RMSE surge (ASF =

+4473.8%) as SOH fades to 80%, whereas parameter-tracking DEKF confines aged RMSE to 0.585% (ASF = +206.3%). (2) *EIS Correlation*: Randles–CPE Nyquist analysis links > 150% R_{ct} expansion to polarization mismatch. (3) *LFP Tracking*: Across flat $LiFePO_4$ plateaus, DEKF maintains 0.870% RMSE by tracking Coulomb gain and ohmic drop without strain sensors. (4) *Pack Inconsistency*: Lumped observers incur 1.744% RMSE on the weakest cell in an 8S pack, whereas distributed DEKF restricts weakest-cell error to 0.337%. (5) *Hybrid PIDKF*: Neural compensation dominates on aged NMC (0.414% RMSE) while hysteresis tracking dominates on aged LFP (0.149% RMSE), executing in 14.20 μs (2, 386 Cortex-M4 cycles).

Acknowledgment and Data Availability

Open-access CALCE and Oxford battery datasets are gratefully acknowledged. Full code pipelines, model parameters, and CMSIS-DSP C kernels are available at https://github.com/lehoang-research/battery-soc-jeecs-benchmark.

A State-Space Matrices and Analytic Jacobians

For 3-state NMC and 4-state LFP models, system matrices are $\mathbf{A}_k^{\text{NMC}} = \text{diag}(1, a_1, a_2)$, $\mathbf{A}_k^{\text{LFP}} = \text{diag}(1, a_1, a_2, a_{h,k})$, $\mathbf{H}_k^{\text{NMC}} = [V'_{\text{ocv}}(\hat{z}_k^-), -1, -1]$, and $\mathbf{H}_k^{\text{LFP}} = [\mathbf{H}_k^{\text{NMC}}, -1]$, with $a_i = e^{-\Delta t/\tau_i}$, $a_{h,k} = e^{-\beta|\Delta z_k|}$, $\mathbf{B}_k = [B_{1,k}, B_{2,k}, B_{3,k}, (0)]^T$, $\mathbf{g}_k = [0, 0, 0, (1 - a_{h,k})M(z_k)\,\text{sgn}(I_k)]^T$, and $\mathbf{C}_{\theta,k} = [-I_k, -\frac{\eta\Delta t I_{k-1}}{3600} V'_{\text{ocv}}(\hat{z}_k^+)]$. Locally ($a_{h,k} \approx a_h$), $|\det(O_{n,k})| = V'_{\text{ocv}}(z_k) \prod_{i=1}^{n-1}(1 - a_i) \prod_{i<j} |a_i - a_j|$; full rank requires $V'_{\text{ocv}} > 0$, $I_k \neq 0$, and non-resonance $a_{h,k} \neq a_1, a_2$. On the plateau (20%–80% SOC), constant saturation $M(z) \equiv M_0 = 30$ mV renders $\mathbf{A}_k^{\text{LFP}}$ block-diagonal with spectrum $(1, a_1, a_2, a_{h,k})$, yielding determinant Eq. (18). Evaluated numerically across $M = 1,370$ sliding windows ($N = 30$ s) over UDDS cycles (80%–100% SOH) under normalization $\tilde{\mathbf{x}} = [z, U_1/0.1, U_2/0.1, V_h/0.03]^T$ and PE ($\|I\|_{[k,k+N]} \geq 0.5$ A), minimum LTV Gramian singular value satisfies $\lambda_{\min}(\mathcal{G}_o) \geq 1.42\times10^{-3}$ at 25°C ($\geq 1.18\times10^{-3}$ at 0°C; $\geq 1.55\times10^{-3}$ at 45°C), empirically confirming uniform complete observability across dynamic drive cycles to complement local analytical rank.

B Discrete Quasi-Sliding Mode Stability and UUB of SMO

Along sliding manifold $S_k = V_{\text{meas},k} - \hat{V}_{t,k}$, one-step dynamics follow $S_{k+1} = S_k + \Delta f_k - \gamma(\hat{z}_k)\,\text{sat}(S_k/\mu)$ with scheduled gain $\gamma(\hat{z}) \equiv \mathbf{H}_k\mathbf{L}(\hat{z}) = l_{\text{smo}}$ ($V'_{\text{ocv}} \geq \alpha_{\min}$) and modeling disturbance $\Delta f_k = -R_0\Delta I_k + [V'_{\text{ocv}}(\xi_{k+1}) - V'_{\text{ocv}}(\xi_k)]e_{z,k} - V'_{\text{ocv}}\Delta Q_k + \sum_{i=1}^{2}[(1-a_i)e_{U_i,k} - d_{U_i,k}] + [(1-a_{h,k})e_{h,k} - \Delta M_k] + (v_{k+1} - v_k)$, where $\Delta I_k = I_{k+1} - I_k$, capacity mismatch satisfies $V'_{\text{ocv}}\Delta Q_k = V'_{\text{ocv}}\frac{\eta\Delta t|I_k|}{3600}|\frac{1}{Q_{\text{act}}} - \frac{1}{\hat{Q}_{\text{act}}}| \leq 0.35$ mV, and v_k denotes bounded AFE noise ($|v_k| \leq V_{\text{AFE}} \leq 0.84$ mV, Arbin ±0.02% F.S.; distinct from filter R_v). Under nominal cycles ($\leq$ 2.5C NMC, $\leq$ 2C LFP; $|\Delta I_k| \leq 0.025$ A): for NMC ($\bar{e}_h = 0$), $|\Delta f_k^{\text{NMC}}| \leq 1.25\text{ mV}(R_0|\Delta I|) + 0.70\text{ mV}(\delta V'_{\text{ocv}}) + 0.35\text{ mV}(\Delta Q) + 0.20\text{ mV(RC)} + 1.68\text{ mV}(2V_{\text{AFE}}) = 4.18\text{ mV} \leq \Phi_d \equiv 0.005$ V; for LFP ($M_0 = 30$ mV, $\bar{e}_h = 60$ mV, $\Delta z \leq 0.00056$), $(1 - a_{h,k})\bar{e}_h \leq (1 - e^{-45\times0.00056}) \times 60\text{ mV} = 1.49$ mV, yielding $|\Delta f_k^{\text{LFP}}| \leq 1.25 + 0.10 + 0.10 + 0.20 + 1.49 + 1.68 = 4.82\text{ mV} \leq \Phi_d \equiv 0.005$ V. At 4C pulses, $|\Delta f_k| \leq \Phi_{d,4C} = 6.16\text{ mV} < 2\mu - l_{\text{smo}} = 15$ mV, preserving reaching decrement $\eta_{\text{reach,4C}} = 8.84\text{ mV} > 0$ and ultimate set $r_{\text{ult,4C}} = 8.21\text{ mV} < \mu$ (Phase 7 faults with $I_{\text{bias}} = 0.25$ A and EMI = 25 mV exceed Φ_d, benchmarking graceful degradation). With candidate $V_k = \frac{1}{2}S_k^2$, increment $\Delta V_k = \frac{1}{2}(S_{k+1} - S_k)(S_{k+1} + S_k)$ satisfies: (1) *Contraction*: Outside $\mathcal{B}_\mu \equiv \{|S| \leq \mu\}$, $\text{sat}(S_k/\mu) = \text{sgn}(S_k)$, yielding $S_{k+1} - S_k = -l_{\text{smo}}\,\text{sgn}(S_k) + \Delta f_k$ and $S_{k+1} + S_k = 2S_k - l_{\text{smo}}\,\text{sgn}(S_k) + \Delta f_k$. Condition $\Phi_d < l_{\text{smo}} < 2\mu - \Phi_d$ ($\Phi_d = 0.005$ V $< l_{\text{smo}} = 0.025$ V < 0.035 V) ensures $\Delta V_k < 0$, ensuring decrement $\eta_{\text{reach}} = \min(l_{\text{smo}} - \Phi_d, 2\mu - l_{\text{smo}} - \Phi_d) = 0.010$ V ($|S_k| - |S_{k+1}| \geq \eta_{\text{reach}}$), entering $\mathcal{B}_\mu$ in $k^* \leq \lceil(|S_0| - \mu)/\eta_{\text{reach}}\rceil$ steps. (2) *Invariance*: Inside $\mathcal{B}_\mu$, $S_{k+1} = (1 - l_{\text{smo}}/\mu)S_k + \Delta f_k$. Since $|1 - l_{\text{smo}}/\mu| = 0.25 < 1$, boundary $\Omega_{\text{ult}} \equiv \{|S| \leq \frac{\Phi_d}{1-|1-l_{\text{smo}}/\mu|} = 6.67\text{ mV}\} \subset \mathcal{B}_\mu$ is positively invariant. (3) *UUB*: Perturbations satisfy $|d_{U_i,k}| \leq (1 - a_i)\tilde{D}_{U_i}$ with $\tilde{D}_{U_i} \equiv R_i(T, \text{SOH})I_{\text{bias}} + \delta R_i I_{\text{RMS}} \leq 3.5$ mV across 0°C–45°C (≤ 2.2 mV at 25°C), yielding tight ISS bounds $|e_{U_i}| \leq \bar{e}_{U_i} \leq 3.5$ mV ($l_2 = l_3 = 0$); LFP hysteresis satisfies $|e_h| \leq 2M_0 \equiv \bar{e}_h = 60$ mV ($l_h = 0$; $\bar{e}_h = 0$ for NMC). By MVT, $|e_{z,k}| \leq (\mu + 2\bar{e}_U + \bar{e}_h + V_{\text{AFE}})/\alpha$. For NMC ($\alpha_{\text{NMC}} \geq 0.40$ V), state error is quantitatively tight ($|e_z| \leq 7.2\%$ across T and SOH; $\leq 6.4\%$ at 25°C). For LFP ($\alpha_{\text{LFP}} = 0.00275$ V), finite UUB $\|\mathbf{e}_x\| < \infty$ is analytically guaranteed, while dynamic hysteresis co-estimation restricts empirical RMSE to 0.941% (fresh) and 0.149% (PIDKF aged).

C ECM Parameter Calibration and Discrete Riccati Covariance Invariance

To facilitate hardware implementation and full reproducibility, Table 10 details the multi-temperature and aging calibration parameters for both chemistry baselines (calibrated from CALCE and Oxford experimental titration procedures).

Table 10 Calibrated 2-RC model parameters across thermal and aging regimes

Parameter	INR20R (NMC)		A123 (LFP)	
	100%	80%	100%	80%
Q_{act}(25°C) (Ah)	2.00	1.60	2.30	1.84
Q_{act}(0°C) (Ah)	1.78	1.42	2.05	1.64
Q_{act}(45°C) (Ah)	2.06	1.65	2.36	1.89
R_0(25°C) (mΩ)	35.0	52.5	22.0	33.0
R_0(0°C) (mΩ)	74.6	111.8	46.9	70.3
R_0(45°C) (mΩ)	24.5	36.8	15.4	23.1
R_1(25°C) (mΩ)	15.0	37.5	12.0	30.0
R_1(0°C) (mΩ)	42.8	106.9	34.2	85.5
R_1(45°C) (mΩ)	9.8	24.5	7.8	19.5
C_1 (F)	1,600	1,200	2,100	1,650
R_2(25°C) (mΩ)	25.0	45.0	18.0	36.0
C_2 (F)	4,500	3,800	6,200	5,100
Hysteresis β	—	—	45.0	45.0
Hysteresis M_0 (mV)	—	—	30.0	30.0

Under the Joseph-stabilized Riccati difference equation:

$$\mathbf{P}_{k+1}^- = \mathbf{A}_k\mathbf{P}_k^+\mathbf{A}_k^T + \mathbf{Q}_k, \tag{C1}$$

$$\mathbf{P}_k^+ = (\mathbf{I} - \mathbf{K}_k\mathbf{H}_k)\mathbf{P}_k^-(\mathbf{I} - \mathbf{K}_k\mathbf{H}_k)^T + \mathbf{K}_k R_{\text{eff},k}\mathbf{K}_k^T, \tag{C2}$$

the discrete algebraic Riccati operator maps the positive definite cone $\mathbb{S}_{++}^n$ into itself. Because $\mathbf{A}_k$ is contractive ($|\lambda_i(\mathbf{A}_k)| \leq 1$), process covariance satisfies $\underline{q}\mathbf{I} \preceq \mathbf{Q}_k \preceq \bar{q}\mathbf{I}$, and innovation covariance satisfies $0 < \underline{r} \leq R_{\text{eff},k} \leq \bar{r} < \infty$, the uniform complete observability (UCO) condition $\mathcal{G}_o(k, k+N) \succeq \alpha_o\mathbf{I}$ establishes that the Riccati mapping is strictly contractive under the Riemannian metric on $\mathbb{S}_{++}^n$. By the discrete Kalman-Bucy theorem, the prior and posterior error covariances are uniformly upper- and lower-bounded for all $k \geq N$:

$$\underline{p}\mathbf{I} \preceq \mathbf{P}_k^+ \preceq \mathbf{P}_k^- \preceq \bar{p}\mathbf{I}, \tag{C3}$$

with explicit bounds $\underline{p} = \underline{q} > 0$ and $\bar{p} = (\alpha_o^{-1} + N\bar{q})\bar{a}^{2N} < \infty$. Consequently, the Kalman gain $\|\mathbf{K}_k\|_2 \leq \bar{p}\|\mathbf{H}_k\|_2/\underline{r} \leq \bar{K} < \infty$ remains uniformly bounded across all operational drive cycles, preventing covariance divergence and validating the UUB stochastic stability theorems in Section 3.7 and Appendix B.

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